

[Draft]

**User's Manual for
Computerized Benefit-Cost Analysis of Electronic Toll Collection System**

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User's Manual – BASIC

This document describes basic use of the Electronic Toll Collection Cost/Benefit Model. It assumes the user has limited familiarity with Excel or a similar spreadsheet as well as with benefit cost analysis.

Copy the file AUTOMAT.xls to a new file name (for instance MYFILE.xls)

Open the new file in Microsoft Excel 97

Go to the page “Input-Start” and fill out any cells you wish to change

The “Input-Start” sheet contains the minimum input required for the model to function. Cells are marked by five different colors:

- White – numbers which must be user input in “Input-Start”.
- Pink - default values that are borrowed from previous studies.
- Blue – interpolated numbers which may be altered by the user.
- Green – calculated number
- Orange - outcomes.

Required input data are categorized in five groups: Traffic Data, Cost Information, Toll Transaction Variables, Travel at the Toll Plaza, and Assumptions about Fuel Consumption and others. All required cells have default values from the Carquinez Bridge study, but you should update these for your location.

Traffic Data - Provide Annual Traffic Volume and Expected ETC Split in base year as well as in year 20; Peak Hour Volume, D factor, and Duration of peak period in the morning peak and the evening peak in base year.

Cost Information - Enter Expected Expenditures in the base year (in each year for one-time costs) in order to operate the toll plaza under the ETC scenario as well as under the no-ETC or baseline scenario. Costs are broken down into One-time Costs and Continuing Costs. Capital Investment is only required only for the ETC alternative.

Toll Transaction Variables - This part collects data about toll transactions. Transaction Times for each type of transaction, Payment Split among Manual users as well as among ETC users under different scenarios, and ETC Account Fact are of interests. You may refer to some in-use toll collection systems to approximate values for these variables.

Travel at the Toll Plaza - Numbers regarding travel profile around the toll plaza are collected in this part. Required information includes Configuration of the Toll Plaza, Prevailing Travel Speed, Mode Split, Average Vehicle Occupancy, and Time Value.

Fuel Consumption and Other Assumptions - How Far we can go by consuming one unit of fuel and How Much we have to pay for a unit fuel are associated with the costs saving in purchasing fuels. The Emission Rate during Acceleration and Idling, and Costs of Air Pollution are tied with environmental concerns. Other numbers like Inflation Rate, Annual Interest Rate, Discount Rate, and Personal Year Assumptions also play important roles in this analysis.

Go to the page “Results Summary” to get Net Present Value of the project, Benefit-Cost Ratio, and the estimated Internal Rate of Return.

Users Manual – DETAILED

This document describes the use of the Electronic Toll Collection Cost/Benefit Model. It assumes the user has limited familiarity with Excel or a similar spreadsheet as well as with benefit cost analysis.

Introduction

Three categories of worksheets are contained in this file:

- *Inputs.* Data which the user needs to input to perform the analysis.
- *Outputs.* Results from the analysis, including the net present (monetary) value of benefits and costs by category (user, agency, society), benefit/cost ratio, and internal rate of return.
- *Calculations.* Intermediate calculations of benefits and costs by category.

The quickest way to apply the model is to learn by doing. Simply follow the instructions and read the footnotes next to the tables on the input sheet. For those who would like to decode the black box, this manual can help understand what is going on inside.

The overview model use procedure is presented in the next section. This is followed by the description of each table and page in the spreadsheet. All tables, including the required input information and calculations, are explained in this part. The case of California's Carquinez Bridge, which is used to illustrate the analysis procedure, can be found in the appendix.

Model Overview

The system is very simple to use at its most basic level.

1. Copy the file AUTOMAT.xls to a new file name (for instance MYFILE.XLS)
2. Open the new file in Microsoft Excel 97
3. Go to the page "*Input-Start*" and fill out all cells requested.¹
4. Go to the page "*Results Summary*" to get Net Present Value of the project, Benefit-Cost Ratio, and the estimated Internal Rate of Return.²

¹ If you have more detailed information than requested on the page "*Input-Start*" Go to "*Input-Detail*"

² For more detailed Results visit the page "*Results-Detail*" to find out Benefit-Cost Valuation item by item for each group: toll agency, users, and community

Input Sheets



Always Start With The Page “*Input-Start*”

You can find two kinds of input sheets in this model. Always start with the “Input-Start” sheet, which contains all input required for the model to function. The additional data on the “Input-Detail” is optional. In the absence of user-supplied data, “Input-Detail” interpolates data from the “Input-Start” sheet. The user should note that the whole analysis is based on the numbers (either default interpolations or user supplied) found on the “Input-Detail” sheet.

The time horizon of the analysis is set as 20 years in this model. The idea of having two input sheets is that you as a user are obliged to provide information in the base year, and sometimes for year 20 as well. For the intervening years, the model can generate numbers on “Input-Detail” based on information you provide on “Input-Start”. If you have your own, better, forecast, change the values on the “Input-Detail” sheet.

The only sheets you as a user have access in this model are the two input sheets. The rest are locked to prevent damage to the model and to ensure consistency between applications. If you are not so sure about default values or you would like to see how sensitive the number is; modify the appropriate cells on the input sheets.

Checked boxes shown at the end of the description in this manual mean that you are able to “Input-Detail” to check out these items and plug in your own numbers rather than rely on those assigned by the model.

Cells are marked by five different colors.

- White – numbers which must be user input in “Input-Start”.
- Pink - default values that are borrowed from previous studies. Blue – interpolated numbers which may be altered by the user.
- Green - number that is calculated based on the numbers you feed in.
- Orange - outcomes.

If you have better information, it is fine to change pink cells, but generally these should be consistent across applications. In other words, numbers in white cells are those which always vary from case to case, whereas those in pink cells are generally the same across cases.

Input-Start

Input All Information Requested in This Sheet

Required input data were categorized to five groups: Traffic Data, Cost Information, Toll Transaction Variables, Travel at the Toll Plaza, and Assumptions about Fuel Consumption and others.

Traffic Data

Here, you are asked to provide Annual Traffic Volume and Expected ETC Split in base year as well as in year 20; Peak Hour Volume, D factor, and Duration of peak period in the morning peak and the evening peak in base year.

- ♦ Annual Traffic Volume (ATV): the number of vehicles going through the toll plaza per year. That is, if only outbound traffic is tolled, only traffic volume in that direction is counted. ☒
- ♦ Expected ETC Split: market share of ETC traffic, assume 0% in base year. ☒
- ♦ Peak Hour Volume (PHV): the vehicles per hour in both directions during the peak period, in veh/hr. Both volumes in morning peak and evening peak are required. ☒
- ♦ D factor: the percentage of total peak hour traffic volume, in both directions, going through the toll plaza. ☒
- ♦ Duration of peak period (T): length of peak period in the morning as well in the evening, in hrs. ☒

Cost Information

All Numbers on This Chart Have to Be in Current Value (Set in Base Year)

Here, you are asked to estimate Expected Expenditures in the base year (in each year for one-time costs) in order to operate the toll plaza under the ETC scenario as well as under the no-ETC or baseline scenario. Costs are broken down into One-time Costs and Continuing Costs. Capital Investment is only required for ETC alternative exclusively.

- ♦ One-time Cost: expenditures that occur only once at a specific point of time. These include factors such as expanding the manual toll plaza in order to meet increasing demand.
- ♦ Continuing Costs: expenditures that occur every year.
 - ✱ Staff. Instead of giving the amount of staff costs directly, you are asked later to provide the cost per unit personal year (PY) and the number of unit PY for information technology and toll collection, respectively. The reason for this is for the purpose of estimating saving in staff needed as introducing ETC system. Therefore, you don't need to enter this number now³.
 - ✱ Operations/Materials: such as the costs of printer paper, ribbons, diskettes, troubleshooting, recovering lost data, and locating parts and service vendors⁴. ☒
 - ✱ Others: costs other than mentioned above, such as the commute ticket booklet contract, and maintenance costs for the existing system. ☒
- ♦ Capital Investment: the amount of capital investment needed to set up ETC system.

Toll Transaction Variables

Numbers regarding toll transaction are gathered in this part. Transaction Times for each type of transaction, Payment Split among Manual users as well as among ETC users under different scenarios, and ETC Account Fact are of interests. You may refer to some in-use toll collection system to attain approximate values for these variables..

- ♦ Transaction Times. Both cash and ticket are accepted in manual channel. While in principle, cash payments can be further distinguished between exact change or not, we assume an average value

³ See Jianling Li, David Gillen, and Joy Dahlgren, An Illustration of Benefit-Cost Method for the ETC System at Carquinez Bridge, PATH Program, Institute for Transportation Studies, University of California, Berkeley, June 1998.

⁴ Ibid.

- for all cash transactions. Here, you do need to give the mean transaction time for cash and ticket, respectively. For ETC lanes, minimum headway is required to estimate the capacity. Since ETC users don't need to stop completely or even slow down (in a properly designed system) when they pass through the toll plaza, transaction time is zero. Times should be presented in seconds and data is required for base year as well as for year 20. These numbers are needed to estimate the capacity of toll plaza, thus the amount of time saved.
- ♦ Payment Split. For the baseline case, only the payment split between cash and ticket is need. While under the ETC scenario, payment split among Manual Users and ETC Users are required. Payment split among Manual Users may vary under ETC scenario since it is said that current ticket users are more likely to switch to ETC lanes rather than those who pay by cash. Data is required for base year as well as for year 20. These numbers are crucial to estimate interest loss of users, and capacity of toll plaza, and thus time saved. ☑
 - ♦ ETC Account Fact: ETC account can be established either by cash & check or credit card. The following numbers in base year are required.
 - ✖ Average Tags per Account. Since several vehicles may share an account, you need to provide average number of transponders issued per account.
 - ✖ Average Usage per Account: total number of ETC transactions divided by the total number of accounts open. Along with Average Tags per Account, the model can estimate the number of accounts open based on projected annual traffic volume.
 - ✖ Deposits. In some cases, a deposit is required to open an account.
 - ✖ Average Balance other than Deposits. Usually, ETC users are asked to maintain certain amount of money in their account. The average balance and deposit cause users losses in interest.

Travel at the Toll Plaza

Numbers regarding travel profile around the toll plaza are collected in this part. Required information includes Configuration of the Toll Plaza, Prevailing Travel Speed, Mode Split among ETC users, Average Vehicle Occupancy, and Time Value for each mode.

- ♦ Configuration of the Toll Plaza. Three items need to be known about the toll plaza:
 - ✖ Total Number of Lanes. The model can analyze the toll plaza up to 21 lanes.
 - ✖ Prevailing Travel Speed. This number refers to normal travel speed on the adjacent highway section, which is not influenced by the existence of the toll plaza, in miles per hour.
- ♦ Mode Split among ETC users. Vehicles are categorized as autos, trucks, and buses.
- ♦ Average Vehicle Occupancy. Average vehicle occupancy for each mode with the information of mode split tells the amount of time saved.
- ♦ Time Value for a User of Each Mode: value per unit time for an auto driver, truck driver, and a bus rider, respectively, in current value per hour.

Fuel Consumption and Other Assumptions

Assumptions about fuel consumption allow us to estimate the reduction in fuel costs and emissions. How Far we can go by consuming one unit of fuel and How Much we have to pay for a unit fuel are associated with the costs saving in purchasing fuels. The Emission Rate during Acceleration and Idling, and Costs of Air Pollution is tied with environmental concerns. Other numbers like Inflation Rate, Annual Interest Rate, Discount Rate, and Personal Year Assumptions also play important roles in this analysis.

- ♦ Assumptions about Fuel Consumption
 - ✖ Average Mileage of Consuming one Gallon of Fuel: how far you can go in average by using one unit of fuel, in miles per gallon.
 - ✖ Fuel Price per Gallon: how much you have to pay in average to get one unit of fuel, in current value per gallon.
 - ✖ Emission Rates during Acceleration. NO_x, HC, and CO are main contributors to harm the environment, in grams per gallon.
 - ✖ Emission Rate during Idling, in grams per gallon.
 - ✖ Cost of Air Pollution, in current value per kilogram of pollutant.

- ♦ Other Assumptions
 - ✖ Inflation Rate.
 - ✖ Annual Interest Rate.
 - ✖ Discount Rate.
 - ✖ Personal Year (PY) Related Assumptions. The model estimates the number of staff needed in terms of units of personal year, and multiplies by the unit cost per PY, to get costs differences in staff between two scenarios.
 - Hours per Unit Personal Year: the amount of hours a full-time employee works per year, in hours per year⁵. Assume no change over time.
 - PY for Information Technology. Staff can be separated into two groups. One is for the computer service support to assist the personnel of the accounting with problems downloading and storing toll data⁶, in units of PYs. Assume no change over time.
 - PY for Program Costs: staff for the toll collection and accounting⁷, in units of PYs. Assume certain relationship between this item and traffic volume.
 - Unit Cost per Personal Year. Assume unit cost per PY doesn't change in real term⁸.

⁵ See Jianling Li, David Gillen, and Joy Dahlgren, An Illustration of Benefit-Cost Method for the ETC System at Carquinez Bridge, PATH Program, Institute for Transportation Studies, University of California, Berkeley, June 1998.

⁶ Ibid.

⁷ Ibid.

⁸ Ibid.



Input-Details



Only Change Numbers in Blue Cells, or Go Back to “Input-Start”

Tables on this page are optional. If you have information regarding items the years between the base year and year 20, and a checked box indicates that you can check it out in “Input-Detail”, this is the place you are looking for. The rationale of this sheet is that the model requires some data for each year to proceed with the analysis. However, users may not have those numbers. Therefore, the model generates the intervening numbers, but it is your responsibility to make reasonable revisions. Thus, reasonable results come out. Following are items that you can find in this sheet and a brief description of each.

I. Traffic Data

Table Ia, Ib, and Ic contain traffic data. All numbers regarding the peak must distinguish between morning and evening peak.

- ♦ Traffic Data-Annual (Ia)
 - ✱ Annual Traffic Volume (ATV). Based on the annual traffic volume in base year and in year 20, which users enter on “Input-Start” sheet, the model generates numbers in between by interpolation, assuming a constant growth rate.
 - ✱ Expected ETC Split. Based on the annual expected ETC split in base year and in year 20, which users enter on “Input-Start” sheet, the model generates numbers in between by interpolation, assuming a constant growth rate.
- ♦ Traffic Data-Peak (Ib-morning peak, Ic-evening peak)
 - ✱ Peak Hour Volume (PHV). The model assumes that the ratio of annual traffic volume to peak hour volume is fixed. The ratio is shown on the column with entry ATV/PHV. Consequently, PHV in year i is estimated by dividing projected annual traffic volume in year i by ATV/PHV.
 - ✱ D factor. Again, this value is assumed constant over time.
 - ✱ Duration of peak period (T). This value is also assumed constant over time.

II. Costs Information

Costs are estimated under ETC scenario as well as under baseline scenario. Operations/Materials costs and other costs are places you should look at.

- ♦ Operations/Materials. Assume the quantity consumed remains constant over time. Thus, only input prices swell in proportion to inflation rate. The amount of expenditures adjusts according to the inflation only. Given the amount of cost in base year, $C_m(0)$, we can estimate costs in year n:

$$NC_m(n) \text{ in year } n = NC_m(n-1) * (1 + I_f)$$

where

I_f : inflation rate

- ♦ Others. Assume the quantity consumed remains constant over time. Thus, the amount of expenditures adjusts according to the inflation only. Given the amount of cost in base year, $C_o(0)$, we can estimate costs in year n:

$$NC_o(n) \text{ in year } n = NC_o(n-1) * (1 + I_f)$$

Toll Transaction Variables

- ♦ Expected Payment Split (Table IV). Based on the payment split in base year and in year 20, which users enter on “Input-Start” sheet, the model generates numbers in between by interpolation, assuming a constant growth rate.

Output Sheets

Two kinds of output sheets are offered in this model: output summary and detailed results. After you complete the input sheets, the model calculates automatically. Just click to the sheets next to the input sheets; and you can find analytical results.

The categories and the valuation of costs and benefits are based on proposed approach suggested by Jianling Li et al. in “An Illustration of Benefit-Cost Method for the ETC System at Carquinez Bridge”. Though some minor adjustments have been made, the framework remains largely the same.

Results Summary

Here, you will find most useful numbers you need: NPV, PVC, PVB, B/C Ratio, and IRR. Moreover, you can also find the overall cost streams as well as benefit streams item by item in this sheet.

- ♦ NPV: net present value. $NPV = PVB - PVC$.
- ♦ PVC: present value of cost streams. Here, costs refer to capital investment needed to set up ETC system exclusively. In other words, this only takes into account capital investment of toll agency.

$$PVC = \sum_{n=0}^{20} \frac{NC_n}{(1 + r_d)^n}$$

where

NC_n : costs streams (including inflation) in year n ,
 r_d : discount rate,
 n : year

- ♦ PVB: present value of monetary benefit streams. Every kind of benefit is converted to equivalent monetary value first and then do the analysis. The sign of benefit may not be always positive. Some sorts of costs are considered as negative benefits.

$$PVB = \sum_{n=0}^{20} \frac{NB_n}{(1 + r_d)^n}$$

where

NB_n : benefits streams (including inflation) in year n ,
 r_d : discount rate,
 n : year

- ♦ B/C Ratio: $B/C = PVB/PVC$. It serves as an indicator of the efficiency of capital investment.
- ♦ IRR: internal rate of return. An estimated rate that makes $NPV = 0$.

$$NPV = \sum_{n=0}^{20} \frac{(NB_n - NC_n)}{(1 + irr)^n} = 0$$



Results-Detail

These tables summarize the estimates of costs and benefits for ETC system cost and benefit streams for each of three groups: Toll Agency, Users, and Community. This enables us to determine who pays most and who benefits most from this sheet.

- ♦ Costs: only ETC capital investment are counted, so all costs are borne by the toll agency. Since in general ETC operating costs are less than Manual operating costs, the difference in costs between the two is classified as a benefit. Note that the use of the difference of costs as a benefit (rather than having a cost category for each set of operating costs) does not affect the net present value, however, it will affect the ratio of benefits to costs.
- ♦ Benefits:
 - ✖ Toll Agency. The toll agency has two kinds of possible benefits: Operating Costs Saving, and Increasing Operational Revenues.
 - Operating Costs Saving, as a result of declining units of PYs in toll collection after introducing ETC system. Please refer to “*Delay*”, “*PY*”, and “*Agency Cost Reduction*” in the section “*Calculation Procedures*” to find out more about this.
 - Increasing Operational Revenues, from interest accumulated in the ETC account. This amount should be equal to users’ interest loss. Please refer to “*Interest Gain*” in the section “*Calculation Procedures*” to know more about this. This value is simply a transfer from users to the agency, and so does not affect the ultimate B/C or NPV calculations.
 - ✖ Users. The benefits to users include the cost reduction in Fuel Consumption, and Time Saving. The incurred Interest Loss due to required minimum balance in ETC accounts is a negative benefit.
 - Cost Reduction in Fuel Consumption: shown in the column “ Δ Fuel Value”. The amount is estimated by the amount of vehicle time-saving, which is a result of a shorter queue, and the elimination of the acceleration/deceleration process multiplied by the unit cost of fuel. Please refer to “*Fuel & Emission*” in the section “*Calculation Procedures*” for details.
 - Time Saving, shown in the column “ Δ Time Value”. The value of passenger time saving is the amount of vehicle time saving weighted by the mode split, average vehicle occupancy of each mode, and unit time value for each passenger of each mode. Please refer to “*Time Value*” in the section “*Calculation Procedures*” to know more about this.
 - Interest Loss, shown in the column “ Δ Interest Gain”. The value of interest loss due to required deposit and minimum balance in ETC account. Please refer to “*Interest Gain*” in the section “*Calculation Procedures*” for additional information.
- ♦ Community: the main benefit to the community is the improvement in the quality of life. ETC system can reduce the fuel consumption, thus emission of pollutants, namely NO_x, HC, and CO. The value of environmental benefit is estimated by multiplying the unit cost to remove pollutants or avoid health damage by the quantity reduction in toxic emission. The numbers are shown in the column “ Δ Env. Value Please refer to “*Fuel & Emission*” in the section “*Calculation Procedures*” to learn more about this.

Calculation Procedures

The procedure to estimate the value of each variable is presented in this section. In this computerized file, each category of costs or benefits run the estimation on one to three sheets. Thus, the procedure is explained by the order of sheets.

- ♦ *Costs estimation is done on the sheet “Agency Cost Reduction”.*
- ♦ *Benefits estimation:*
 - ✖ *Agency:*
 - Costs Saving: “*Agency Cost Reduction*”
 - Increasing Operation Revenues: “*Interest Gain*”
 - ✖ *Users:*
 - Time Saving: “*Time Value*”
 - Fuel Value: “*Fuel & Emission*”
 - ✖ *Community*
 - Environmental Value: “*Fuel & Emission*”



Agency Cost Reduction

On this sheet, the agency's costs and benefits are estimated as capital investment and costs saving.

- ♦ Capital Investment. Capital investment is required input to tell how much the agency has to pay to set up the ETC system. The numbers shown on this sheet are slightly different from the input sheet due to taking into account the inflation effect. Therefore, the amount here is not current value but nominal value.
- ♦ Reduction in Operating Costs. The costs for operations/materials and others are the numbers copied from "Input-Details". The key issue is to estimate staff costs in terms of PYs by the following steps: finding out the number of ETC lanes in year n first, calculating the units of PY needed for toll collections, and then multiplying total units of PY by unit cost of PY.
 - ✱ The number of ETC Lanes in year n: see estimation on "Delay".
 - ✱ Units of PY needed for toll collections: see estimation on "PY".
 - ✱ Total staff costs:

$$SC_n = UCPY * (PYI + PYT_n) * (1 + I_f)^n$$

where

SC_n : staff cost in year n.

UCPY : cost of unit personal year in base year, in \$.

PYI : units of PY needed for information technology, assume fixed over time.

PYT_n : units of PY needed for toll collection in year n, varying with traffic volume.

- ✱ Total Operating Costs in Year n (TC_n):

$$TC_n = OTC_n + SC_n + MOC_n + OC_n$$

where

OTC_n : One-Time Costs incurred at year n.

MOC_n : Operations/Materials Costs in year n.

OC_n : Other costs in year n.

- ✱ After estimating operating costs for baseline and ETC scenario, the difference between two scenarios are Agency Cost Reduction.

Delay

The main task in this sheet is to find the optimal number of ETC lanes given traffic level, transaction times, and total number of lanes. The optimal number of ETC lanes is the one that minimizes total delay in the system. Adding a constraint that average delay in ETC lanes should be always less than that in manual lanes, the suggested number of ETC lanes are used for later estimation. The constraint is to encourage usage of ETC lanes in the long run by ensuring that there is an ETC time savings.

One important assumption here is that only delay within the peak period is regarded as significant. Though minor off-peak delay may occur occasionally, effect can be neglected compared to severe condition within the peak period.

In general, delay can be separated into two kinds: random delay and stop delay.

- ♦ Random Delay (RD). The formula employed to estimate random delay is called generalized delay model, suggested by Ling, L. Fambro, D. B. & Roupail, N. M⁹, for new Highway Capacity Manual.

$$RD_i = 900T \left[(\rho_i - 1) + \sqrt{(\rho_i - 1)^2 + \frac{8kI\rho_i}{Tm_i\mu_i}} \right]$$

where

- RD_i : average random delay, in sec. $i = 1$ for manual, and $i = 2$ for ETC lanes.
- T : duration of peak hour, T_m for morning and T_e for the evening, in hrs.
- μ_i : capacity of one lane, $i = 1$ for manual, and $i = 2$ for ETC lane, in veh/hr.
- m_i : number of lanes, $i = 1$ for manual, and $i = 2$ for ETC lane.
- λ_i : arrival rate during the peak period, $i = 1$ for manual, and $i = 2$ for ETC lane, in veh/hr.
- ρ_i : degree of saturation, $\rho_i = \frac{\lambda_i}{m_i\mu_i}$
- k : parameter for given arrival and service distribution.
 $k = 1$ for Poisson arrival and exponential service time.
- I : ratio of variance over mean for arrivals. $I = 1$ for Poisson Arrival Process.

According to previous studies, Poisson arrival and exponential service time distribution are considered reasonable assumptions^{10,11,12,13}. Hence, the formula can be simplified to:

$$RD_i = 900T \left[(\rho_i - 1) + \sqrt{(\rho_i - 1)^2 + \frac{8\rho_i}{Tm_i\mu_i}} \right]$$

⁹ Li, Jing, Nagui M. Roupail, and Rahmi AkHelik. Overflow Delay Estimation for a Simple Intersection with Fully Actuated Signal Control. *Transportation Research Record 1457*, TRB, National Research Council, Washington, D.C., 1985, pp.73-81.

¹⁰ Feng-Bor Lin & Cheng-Wei Su. "Level-of-Service Analysis of Toll Plazas on Freeway Main Lines." *Journal of Transportation Engineering*, Vol. 120, No. 2, March/April, 1994, pp. 246-263.

¹¹ H.M. Al-Deek, A.E. Radwan, A.A. Mohammed & J.G. Klodzinski. "Evaluation the Improvements in Traffic Operations at a Real-Life Toll Plaza with Electronic Toll Collection." *ITS Journal*, Vol. 3(3), 1996, pp. 205-223.

¹² Mark Robinson & Michel Van Aerde. "Examining the Delay and Environmental Impacts of Toll Plaza." In *1995 Vehicle Navigation & Information Systems conference Proceedings - 6th International VNIS*, July 30 - August 2, 1995, Washington State Convention and Trade Center, Seattle, Washington, USA.

¹³ T. Hugh Woo & Lester A. Hoel. "An Investigation of Toll Plaza Capacity and Level of Service." Virginia Transportation Research Council, 1991.

- ♦ Average Stop Delay (SD_i). $1/\mu_i$ represents the mean transaction time. For the manual toll collection channels, it is a transaction time weighted by payment split. For ETC, users don't need to stop down to pay toll; therefore, transaction time is zero, and capacity is estimated by mean headway. That is, stopped delay for manual users is equal to $1/\mu_1$. Stopped delay for ETC users is zero.

- ♦ Average Delay per user per day (AD_m) in year n (sec):

$$\forall D^m = \forall D^m + \forall D^m$$

$$\Delta DD^m = (\forall D^{m,m} \cdot \gamma^{m,m} + \forall D^{m,m} \cdot \gamma^{m,m}) \cdot \Delta^m \setminus 3600$$

- ♦ Total Delay during the morning peak in the System (TDD_{nm}) per day (hr): Estimate total delay during the evening peak by the same procedure.
- ♦ Optimal number of ETC lanes. The model estimates total delay in the system for each possible combination of ETC and manual lanes. The combination that gives minimum value of total delay is chosen. However, to make sure average delay suffered by ETC users is less than Manual users; the *suggested* number of lanes is equal to optimal number of lanes plus one when the above condition is violated.
- ♦ Total Delay in year n (TD_n): subscript b for baseline and c for ETC scenario, in hrs. Summation of total delay in the evening and in the morning per day multiplied by the total number of weekdays per year gives us the total delay per year under ETC as well as baseline scenario.



PY

The main task on this sheet is to find the units of Person Years needed for Toll Collection in year n . The model assumes that current units of Person Years are well balanced with traffic level. Notice that, during the peak period, the toll plaza should open all the channels to meet the demand. Based on the length of the peak period and the total number of manual lanes, the model estimates how many units of PY are needed during the peak period for a year. The remaining person years are thus for the off-peak period.

The schedule of number of manual channels open are unknown. Suppose that units of PY needed for off-peak period is linear to off-peak traffic level. Off-peak traffic volume can be estimated by subtracting from the projected annual traffic volume the peak traffic per year $((PHV_m * T_m + PHV_e * T_e) * \text{numbers of weekdays per year})$. Thus, units of PY in year n for off-peak period can be approximated.

On the other hand, suggested number of ETC lanes from “*Delay*” gives the number of manual channels open during the peak period every day. Therefore, we can get the required units of PY for both peak period and off-peak period.



Interest Gain

The main task on this sheet is to calculate the interest accrued from the toll agency's perspective, or the interest loss from the users' point of view, as a result of required minimum balance and deposit in ETC account. There is no net costs incurred or benefits generated.

Estimating the interest rate generated by the amount in the ETC account, following procedure are chosen.

- ♦ Average Balance for Payment Type j in year n (AB_{jn}).

$$AB_{jn} = ATAG_{jn} \cdot DT_{jn} + ABXT_{jn}$$

where

$ATAG_{jn}$: average tags per account for payment type j in year n.

DT_{jn} : required amount of deposit for transponder for payment type j in year n.

$ABXT_{jn}$: average balance other than deposit per account for payment type j in year n.

- ♦ Number of ETC Account Set up with Payment Type j in year n (AC_{jn}).

$$AC_{jn} = ET_n \cdot SP_{jn} / AU_{jn}$$

where

ET_n : total number of ETC transaction in year n. $ET_n = ATV_n \cdot \text{expected ETC Split}$.

SP_{jn} : split of payment type j among ETC users in year n.

AU_{jn} : average usage per account for payment type j in year n.

- ♦ The Value of Interest Generated per year by ETC Account in year n (I_n).

$$I_n = \sum_j (AC_{jn} \cdot AB_{jn}) \cdot Iv$$

where

Iv : annual interest rate.

For convenience, ATAG, DT, ABXT, SP, AU are assumed fixed across the analysis period.



Time Value

The value of time saved is estimated on this sheet. The time savings come from reduced acceleration/deceleration process and random/stop delay. In the manual lanes, users have to slow down from prevailing travel speed to stop completely for toll transaction, and then speed up to normal speed. To value time saved, vehicle time saving and passenger time saving have to be calculate first. Along with the appropriate number assigned to the value of time, the monetary value of time saving is obtained.

- ♦ Total Time spent in Acceleration/Deceleration Process for Manual Users in year n (T_{amn}/T_{dmn})
where

$$T_{dmn} = (ATV_n - ET_n) \cdot D_d / V_d$$

D_a : the length of ramp leading from the toll plaza, in miles.

D_d : the length of ramp prior to toll plaza, in miles.

V_n : prevailing travel speed on adjacent highway section, in mph.

V_a : average travel speed during the acceleration process, in mph.

For manual users, assumed constant acceleration rate; thus, $V_a = 0.5 \cdot V_n$

V_d : average travel speed during the deceleration process, in mph.

For manual users, assumed constant deceleration rate; thus, $V_d = 0.5 \cdot V_n$

- ♦ Total Time spent in Acceleration/Deceleration Process for ETC Users in year n (T_{aen}/T_{den})
- ♦ Total Time Saving in year n (TS_n).

$$TS_n = (T_{abn} - T_{acn}) + (T_{den} - T_{aen})$$

where

b, c : denote baseline and ETC scenario, respectively.

a, d : denote acceleration and deceleration process, respectively.

- ♦ Factor of Time Value in year n (F_n).
where

$$F_n = (1 + I_f)^n \cdot \sum_k U_k \cdot VOC_k \cdot SU_k$$

U_k : value of time per hour for passenger using mode k in base year.

VOC_k : average vehicle occupancy for mode k in base year.

SU_k : mode split among ETC users for mode k in base year.

- ♦ Value of Time Saving in year n (US_n).

$$US_n = TS_n \cdot F_n$$

For the convenience of analysis, we assume that the real value of time, the mode split, and the average vehicle occupancy won't change significantly over the analysis period.



Fuel & Emission

Fuel reduction in terms of purchasing costs and pollutant emission are estimated on this sheet. Three main pollutants are of interests: NOx, HC, and CO.

- ♦ Fuel Value in year n (FV_n). This refers to the cost reduction in fuel purchases as a result of the reduction in travel time and thus, fuel consumption. In general, engines need more fuel during acceleration than otherwise. Therefore, only the reduction of fuel consumption during the acceleration process is taken into account. We exclude the fuel savings due to reduced idling associated with ETC, which we expect to be smaller. The amount of cost reduction can simply obtain as following:

$$FV_n = (T_{abn} - T_{acn}) \cdot FG_a \cdot UFC \cdot (1 + I_f)^n$$

where

FG_n : average fuel consumption per hour in year n, in gallons.

UFC_n : unit cost per gallon in base year.

- ♦ Environmental Value in year n (EV_n). The main benefits to the environment are emission reductions of NOx, HC, and CO during the idling and acceleration process.

- ✖ Emission Reduction of Pollutant from Idling event in year n (ER_{pn}), in kgs.

$$ER_{pn} = T_{idle,n} \cdot E_{idle,p} \cdot 60 / 1000$$

where

$T_{idle,n}$: time saved in the idling event in year n, in hrs.

It is equal to differences in delay under baseline and ETC scenario.

$E_{idle,p}$: emission rate for pollutant type p during the idling event, in grams per minute.

- ✖ Emission Reduction of Pollutant p from Acceleration Process in year n ($ER_{acc,p,n}$), in kgs.

$$ER_{acc,p,n} = (T_{abn} - T_{acn}) \cdot FG_a \cdot E_{acc,p} / 1000$$

where

$E_{acc,p}$: emission rate for pollutant type p during the acceleration, in grams per gallon.

Appendix: Example

The data at Carquinez Bridge was used to illustrate step by step how this model works. All input data was retrieved from the previous Path report by Li, J., Gillen, D., & Dahlgren, J.¹ Some changes had been made, including the time horizon, costs/benefits categories, the way to estimate required units of personal years in the future, etc. Hence, absolute value of benefits and costs are different in two models.

Getting Start-“Input-Start”

After copying the file and save it as “Carquinez.xls”, click on the sheet named “*Input-Start*”. The leftmost chart you would see on the screen is shown as Table 1. This chart gives you a quick guide to use this model.

I. Traffic Data

After a quick look, move to the table right next to instruction chart, entitled “TRAFFIC DATA”. Check the definition of information required, and then fill out white cells. Sample input is shown as Table 2.

1. Annual Traffic Volume in base year. According to Li’s report, 19,016,173 vehicles pass through the toll plaza at Carquinez Bridge in year 95/96.
2. Annual Traffic Volume in year 20. According to Li’s report, expected growth rate of annual traffic volume is 3%. Thus, about 20m traffic will pass through toll plaza at Carquinez Bridge in year 20.
3. ETC Split in base year: 0% since no ETC option available.
4. ETC Split in year 20. According to Li’s report, 6% of ETC traffic is expected in year 1, 15% in year 2, and 5% annual growth rate afterwards. However, certain percentage of traffic should be kept for manual users since some people may not establish ETC account if they use Carquinez Bridge only once a while. Therefore, assume the market share of ETC traffic is 98% in year 20.
5. Peak Hour Volume during the morning peak (evening) in base year. According to Caltran², PHV is 7,700 (8,200).
6. Duration of Peak Period in the morning (evening): assumed 2 (2) hrs per day.

¹ See Jianling Li, David Gillen, and Joy Dahlgren, An Illustration of Benefit-Cost Method for the ETC System at Carquinez Bridge, PATH Program, Institute for Transportation Studies, University of California, Berkeley, June 1998.

² Source: Caltrans on the web.

Table 3: Baseline Costs Estimation⁴

	FY 95/96		FY 96/97		FY 97/98		FY 98/99		FY 99/00		FY 00/01		FY 01/02		FY 02/03		FY 03/04		FY 04/05		FY 05/06		TOTAL
	PYs	Amts	PYs	Amts	PYs	Amts	PYs	Amts	PYs	Amts	PYs	Amts	PYs	Amts	PYs	Amts	PYs	Amts	PYs	Amts	PYs	Amts	Amts
Continuing Existing Costs:																							
<i>* information technology costs:</i>																							
Staff	0.11	\$7,222	0.11	\$7,439	0.11	\$7,662	0.11	\$7,892	0.11	\$8,129	0.11	\$8,373	0.11	\$8,624	0.11	\$8,882	0.11	\$9,149	0.11	\$9,423		\$9,706	\$92,501
hardware/software		\$8,356		\$9,390		\$10,587		\$11,975		\$13,589		\$15,472		\$17,673		\$20,252		\$23,280		\$26,841		\$31,038	\$188,454
<i>total information technology cost</i>		\$15,578		\$16,829		\$18,249		\$19,867		\$21,718		\$23,845		\$26,297		\$29,134		\$32,428		\$36,264		\$40,744	\$280,955
<i>* program costs:</i>																							
Staff	61.5	\$3,997,500	61.5	\$4,117,425	61.5	\$4,240,948	61.5	\$4,368,176	61.5	\$4,499,221	61.5	\$4,634,198	61.5	\$4,773,224	61.5	\$4,916,421	61.5	\$5,063,913	61.5	\$5,215,831	61.5	\$5,372,306	\$51,199,163
Other		\$297,120		\$302,720		\$308,800		\$315,200		\$322,240		\$329,760		\$339,653		\$349,842		\$360,338		\$371,148		\$382,282	\$3,679,103
<i>total program costs</i>		\$4,294,620		\$4,420,145		\$4,549,748		\$4,683,376		\$4,821,461		\$4,963,958		\$5,112,877		\$5,266,263		\$5,424,251		\$5,586,979		\$5,754,588	\$54,878,266
Total Continuing Existing Costs:		\$4,310,198		\$4,436,974		\$4,567,997		\$4,703,243		\$4,843,180		\$4,987,803		\$5,139,174		\$5,295,398		\$5,456,680		\$5,623,243		\$5,795,332	\$55,159,221
Total Costs		\$4,310,198		\$4,436,974		\$4,567,997		\$4,703,243		\$4,843,180		\$4,987,803		\$5,139,174		\$5,295,398		\$5,456,680		\$5,623,243		\$5,795,332	\$55,159,221

⁴ Ibid.

Table 4: Costs Estimation for ETC Alternative

	FY 95/96		FY 96/97		FY 97/98		FY 98/99		FY 99/00		FY 00/01		FY 01/02		FY 02/03		FY 03/04		FY 04/05		FY 05/06		TOTAL
	PYs	Amts	PYs	Amts	PYs	Amts	PYs	Amts	PYs	Amts	PYs	Amts	PYs	Amts	PYs	Amts	PYs	Amts	PYs	Amts	PYs	Amts	Amts
Process Control Costs																							
In-lane subsystems (without ETC)		\$786,560		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0	\$786,560
ETC addition to in-lane subsystems		\$1,113,265		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0	\$1,113,265
subtotal		\$1,899,824		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0	\$1,899,824
contingencies 5%		\$94,991		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0	\$94,991
Total Process Control Costs		\$1,994,816		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0	\$1,994,816
Data Processing Costs																							
<i>* One-Time costs:</i>																							
staff		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0	\$0
hardware/software		\$1,014,524		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0	\$1,014,524
data center services		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0	\$0
contract services		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0	\$0
agency facilities		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0	\$0
other		\$0		\$0		\$284,813		\$448,581		\$273,801		\$289,568		\$306,036		\$323,231		\$341,183		\$359,920		\$379,475	\$3,006,608
<i>Total one-time costs</i>		\$1,014,524		\$0		\$284,813		\$448,581		\$273,801		\$289,568		\$306,036		\$323,231		\$341,183		\$359,920		\$379,475	\$4,021,132
<i>* Continuing costs:</i>																							
staff	0.11	\$7,222		\$7,439	0.11	\$7,662	0.11	\$7,892	0.11	\$8,129	0.11	\$8,373	0.00	\$8,624	0.11	\$8,882	0.11	\$9,149	0.11	\$9,423	0.9	\$9,706	\$92,501
hardware/software		\$8,356		\$9,390		\$3,600		\$3,780		\$3,969		\$4,167		\$4,376		\$4,595		\$4,824		\$5,066		\$5,319	\$57,442
data center services		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0	\$0
contract services		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0	\$0
agency facilities		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0	\$0
other		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0	\$0
<i>Total continuing costs</i>		\$15,578		\$16,829		\$11,262		\$11,672		\$12,098		\$12,540		\$13,000		\$13,477		\$13,973		\$14,489		\$15,025	\$149,943
Total Data Processing Costs		\$1,030,102		\$16,829		\$296,075		\$460,253		\$285,899		\$302,108		\$319,036		\$336,708		\$355,156		\$374,409		\$394,500	\$4,171,075
Continuing Existing Costs:																							
<i>* information technology costs:</i>																							
staff		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0	\$0
hardware/software		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0	\$0
<i>total information technology cost</i>		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0		\$0	\$0
<i>* program costs:</i>																							
staff	61.5	\$3,997,500	61.5	\$4,117,425	61	\$4,213,364	60.4	\$4,290,046	59.9	\$4,382,169	59.5	\$4,483,492	59	\$4,579,191	58.5	\$4,676,595	58	\$4,775,723	57.4	\$4,868,109	56.8	\$4,961,739	\$49,345,354
other		\$301,760		\$325,760		\$122,080		\$133,760		\$133,920		\$134,080		\$138,102		\$142,245		\$146,513		\$150,908		\$155,435	\$1,884,564
<i>total program costs</i>		\$4,299,260		\$4,443,185		\$4,335,444		\$4,423,806		\$4,516,089		\$4,617,572		\$4,717,293		\$4,818,841		\$4,922,236		\$5,019,017		\$5,117,175	\$51,229,918
Total Continuing Existing Costs:		\$4,299,260		\$4,443,185		\$4,335,444		\$4,423,806		\$4,516,089		\$4,617,572		\$4,717,293		\$4,818,841		\$4,922,236		\$5,019,017		\$5,117,175	\$51,229,918
Total Costs		\$7,324,178		\$4,460,014		\$4,631,519		\$4,884,059		\$4,801,987		\$4,919,680		\$5,036,328		\$5,155,549		\$5,277,392		\$5,393,426		\$5,511,675	\$57,395,809

Table 1: Instruction on “*Input-Start*”

0.

>> INSTRUCTION <<

This spreadsheet model is designed exclusively for benefit-cost evaluation of Electronic Toll Collection system.

Where to Start?

Please start with sheet "**Input-Start**" and fill out all required cells. You may check "**Input-Detail**" and make some changes based on your own knowledge, or go to the "**Results Summary**" and "**Results-Detail**" directly to find out the analytical results.
Notice that: always complete this sheet first.

Other sheets in this files are for the sake of estimation. You may want to have a Glance and get a general idea of how to get these numbers. You can refer to Footnotes right below the spreadsheet which provide formula and assumptions Employed to proceed the calculation.

❑ SHEET "Input-Start"

In this Sheet, you are asked to give the basic information required for Benefit-Cost analysis.

Basically, you have to provide the data for Base Year as well as for Year 20.

Then, the model estimates the numbers in between automatically by linear interpolation.

If you have more data regarding information required other than base in year and in year 20,

Enter the basic data required in this sheet first;

Then, go to the sheet "Input-Detail" to change the number shown in blue cells.

❑ SHEET "Input-Detail"

In this Sheet, you you may change the numbers estimated automatically by this model based on the numbers you enter in "Input-Start".

❑ SHEET "Results Summary"

In this Sheet, , overall analytical results are presented as well as benefit-cost ratio and Estimated internal interest rate.

❑ SHEET "Results-Detail"

In this Sheet, benefit-cost analyses are shown for Toll Agency, Users, and Community. respectively.

Table 2: Traffic Data on “Input-Start”

I. TRAFFIC DATA

INSTRUCTION:

In this section, You are asked to give information about *annual traffic volume, peak hour volume, D factor and duration of peak period and ETC Split* both in base year and in year 20 or one of them.

If you have information in between, go to sheet "Input-Detail" table Ia, Ib, and 1c after completing this table.

Required Inputs - white cells

a) Annual Traffic Volume in Base Year as well as in Year 20:

"Annual Traffic Volume" here is defined as the number of traffic goes through the Toll Plaza per year.

b) Expected ETC Split in Base Year as well as in Year 20

c) Peak Hour Volume (PHV) in Base Year.

"Peak Hour Volume" here is defined as traffic volume in both direction during the peak period.

d) Peak Hour Duration (T) in Base Year

e) D factor (D) in Base Year - directional split

Estimated Annual Traffic Volume (ATV)

	<i>Base Year</i>	<i>Year 20</i>
<i>Traffic Volume</i>	19,016,173	30,000,000
<i>ETC Split</i>	0.0%	98.0%

Peak Hour Statistics in Base Year

	<i>Morning</i>	<i>Evening</i>
<i>PHV (veh/hr)</i>	7,700	8,200
<i>T (hr)</i>	2	2
<i>D factor</i>	0.3	0.7

7. D factor during the morning (evening peak): assume 0.3 (0.7). Only outbound traffic is tolled at Carquinez Bridge. Therefore, peak traffic during a day at the toll plaza occurs in the evening.

II. Costs Information

Next one is “Costs Information” for ETC and baseline, respectively. Check the definition of information required, and then fill out white cells.

Costs have to break down into capital investment, one-time operating costs, costs of staff, cost of operations/materials, and other costs. Costs for ETC alternative and baseline in 10 years from Li’s report are shown on Table 3 and Table 4.

All purchases of equipment to set up ETC system are categorized as capital investment. On Table 4, total process control costs and one-time costs in base year are counted as capital investment in this sense.

Set 95/96 as base year. Item hardware/soft is categorized as operations/materials. Based on Li’s report, no one-time costs incurred in baseline case, cost of operations/materials is \$8,356, and other cost is \$297,120. Similarly observation can be get from Table 4. Table 5 and Table 6 shows filled costs information in the “Input-Start” sheet. Purchase of transponders, and the amount is proportional to new transponder issued in that year is the only source of one-time costs under ETC scenario. Mean unit cost of transponder, 38.67 is used to estimate costs in each year with projected annual traffic volume and ETC split.

III. Fact of Toll Transaction

Sample input is shown as Table 7.

1. Transaction Time: 7 sec for cash³, 4.5 sec for ticket users. Capacity of a ETC lane is about 1,500, thus, average headway is 2.4 secs.
2. Payment Split. Suppose that ticket users are more likely to switch to ETC users. Payment split for manual lanes under ETC scenario, thus, different from under baseline. Numbers are from Li’s report.
3. ETC Account Fact. Numbers are from Li’s report.

IV. Travel at the Toll Plaza

Sample input is shown as Table 8.

1. Total Number of Lanes: 12 at the Toll Plaza at Carquinez Bridge.

³ Users who pay by cash may have exact change or not. According to Mitsuru Saito in “Effect of Branch and Electronic Toll Collections on Delay and User Costs” indicated that mean transaction time without change is about 5.2 sec, whereas mean transaction time with change required is about 9 to 10. Therefore, assume 7 sec for cash users here.

Table 5: Costs Information for Baseline on “Input-Start”

IIa. COSTS INFORMATION--Baseline (No ETC)

INSTRUCTION:

In this section, You are asked to give information about costs for baseline **operations**. Costs are broken down into two categories: one-time costs and **continuing costs**. All the costs are presented in current value.

Go to sheet "Input-Detail" table IIa after completing this table if you have detailed information regarding baseline costs.

Required Inputs - white cells

- a) One-Time Costs: Costs which occur only once at a specific point of time.
enter "0" if no costs incurred.
- b) Continuing Costs: Costs incurred every year
 - 1) Operations/Materials Costs
 - 2) Others

Staff Costs is estimated automatically by given units of personal year needed for toll collection and information technology, and costs per personal year.

Year	One-Time Costs	Continuing Costs:		
		Staff	Operations/Materials	Other
<i>Base</i>	\$0		\$8,356	\$297,120
<i>1</i>	\$0			
<i>2</i>	\$0			
<i>3</i>	\$0			
<i>4</i>	\$0			
<i>5</i>	\$0			
<i>6</i>	\$0			
<i>7</i>	\$0			
<i>8</i>	\$0			
<i>9</i>	\$0			
<i>10</i>	\$0			
<i>11</i>	\$0			
<i>12</i>	\$0			
<i>13</i>	\$0			
<i>14</i>	\$0			
<i>15</i>	\$0			
<i>16</i>	\$0			
<i>17</i>	\$0			
<i>18</i>	\$0			
<i>19</i>	\$0			
<i>20</i>	\$0			

Table 6: Costs Information for ETC Alternative on sheet “Input-Start”

IIb. COSTS INFORMATION--ETC Alternative

INSTRUCTION:

In this section, You are asked to give information about costs under ETC scenario. Costs are broken down into two categories: capital investment and operating costs. All the costs are presented in current value.

If you have more information, go to sheet "Input-Detail" table IIb after completing this table.

Required Inputs - white cells

- a) Capital Investment: Initial investment needed to implement ETC project
- b) Operating Costs: Costs in order to operate the system under ETC scenario
 - 1) One-time costs: costs which occur only once at a specific point of time
 - 2) Operations/Materials Costs
 - 3) Others

Staff Costs is estimated automatically by given units of personal year needed for toll collection and information technology, and costs per personal year.

Year	Capital Investment	Operating Costs			
		One-Time Costs	Staff	Operations/Materials	Other
<i>Base</i>	\$3,009,340	\$0		\$8,356	\$301,760
<i>1</i>	\$0	\$231,886			
<i>2</i>	\$0	\$244,904			
<i>3</i>	\$0	\$257,922			
<i>4</i>	\$0	\$270,940			
<i>5</i>	\$0	\$283,958			
<i>6</i>	\$0	\$296,976			
<i>7</i>	\$0	\$309,994			
<i>8</i>	\$0	\$323,012			
<i>9</i>	\$0	\$336,029			
<i>10</i>	\$0	\$349,047			
<i>11</i>	\$0	\$362,065			
<i>12</i>	\$0	\$375,083			
<i>13</i>	\$0	\$388,101			
<i>14</i>	\$0	\$401,119			
<i>15</i>	\$0	\$414,137			
<i>16</i>	\$0	\$427,155			
<i>17</i>	\$0	\$440,173			
<i>18</i>	\$0	\$453,190			
<i>19</i>	\$0	\$466,208			
<i>20</i>	\$0	\$479,226			

Table 7: Fact of Toll Transaction on “Input-Start”

III. FACT OF TOLL TRANSACTION

INSTRUCTION:

Required Inputs - white cells

If you have more information about payment split other base year and year 20, go to Table IV, expected payment split, in sheet "Input-Details" after completing this table.

Default Values - pink cells

The values shown in the pink cells are the default number we used previously to run the model, please change it if you have different numbers.

- a) Avg. Tags per Account: The ratio of the number of transponders issued over the number of accounts open. Several cars may share one account.
- b) Avg. Usage per Year: Total number of transactions divided by the total number of accounts open.
- c) Deposits: This number refers to the required deposits for transponders.

<u>Transaction Time (sec)</u>		
Cash		7
Ticket		4.5
ETC (headway)		2.4
<u>Payment Split</u>		
<u>Manual Lane-Baseline</u>	<i>Base Year</i>	<i>Year 20</i>
Cash	83%	83%
Ticket	17%	17%
<u>Manual Lane-ETC Alternative</u>		
Cash	83%	100%
Ticket	17%	0%
<u>ETC Lane-ETC Alternative</u>		
Cash, Checks	64%	64%
Credit Card	36%	36%
<u>ETC Account Fact</u>		
<u>Cash & Checks Account</u>	<i>Base Year</i>	
Avg. Tags per Account	1.35	
Avg. Usage per Year	160	
Deposits	\$30	
Avg. Balance other than Deposit	\$20	
<u>Credit Card Account</u>	<i>Base Year</i>	
Avg. Tags per Account	1.35	
Avg. Usage per Year	160	
Deposits	\$0	
Avg. Balance other than Deposit	\$20	

2. Ramp prior to the Toll Plaza: 0.2 miles. It is the distance between the point that drivers start to slow down from the prevailing travel speed to the booth of the plaza.
3. Ramp leading from the Toll Plaza: 0.2 miles. It is the distance between the toll booth to the point that drivers accelerate to the prevailing travel speed after transaction.
4. Prevailing Travel Speed: 55 mph. It is the normal travel speed on adjacent highway section that is not affected by the existence of the toll plaza.
5. Mode Split among ETC Users: 94.76% in auto, 5.11% in trucks, and 0.13% in buses.
6. Average Vehicle Occupancy: 1.8 for auto, 1.1 for truck, and 20 for bus.
7. Time Value of a User for each mode: \$12.75 for auto, \$33.41 for truck, and \$12.75 per bus rider.

V. Fuel Consumption and other Assumptions

Sample input is shown as Table 9. All the numbers on this chart are from Li's report.

Now, we have finished "Input-Start". Let's go to "Input-Detail" to feed in more information we have on hand.

Input-Detail

In this sheet, instead using numbers estimated by the model automatically, we would rather use our own forecast. Data about Traffic Volume, Cost information, and expected payment split are places we should look at.

I. Traffic Data - Annual

Sample input is shown as Table 10. Here, we replaced expected ETC split by our own assumptions mentioned early.

II. Traffic Data – Morning/Evening Peak

Sample input is shown as Table 11. The model assumes that duration of peak hours, directional split and ratio between annual travel volume and peak hour volume don't change over time. We keep those assumptions, thus without any change on these two tables.

III. Costs Information

Sample input is shown as Table 12 & Table 13. Note that the value shown in this table is not current value but nominal value (amount of money in year n). Therefore, the amount of capital investment and one-time costs is different from we input on "Input-Start".

Operations/Material Costs up to FY 00/01 and Other Costs up to FY05/06 had been estimated in Caltrans study. Constant annual growth rate of five percent afterwards was assumed in Li's report. Based on these data and assumptions, we estimated the costs up to year 20.

Table 8: Travel at Toll Plaza on “Input-Start”

IV. TRAVEL AT THE TOLL PLAZA

INSTRUCTION:

In this section, You are asked to give information about travel fact at the toll plaza

Required Inputs - white cells

a) Ramp Distance prior to (leading from) the Toll Plaza: The distance that a vehicle starts to slow down from the normal speed (to accelerate from complete stop).

b) Prevailing Travel Speed: The travel speed in the adjacent highway sections that is not influenced by the existence of the toll plaza.

Default Values - pink cells

The numbers in **pink cells** are used previously to run the model, please update it if your case has different values .

Configuration of the Toll Plaza

Total Number of Lanes (max 22)	12
Ramp Distance prior to the Toll Plaza (miles)	0.2
Ramp Distance leading from the toll Plaza (miles)	0.2

Prevailing Travel Speed (mph)

55

Mode Split among ETC Users

Auto	94.76%
Truck	5.11%
Bus	0.13%

Average Vehicle Occupancy

Auto	1.8
Truck	1.1
Bus	20

Time Value of a User for Each Mode (\$/hr)

Auto	\$12.75
Truck	\$33.41
Bus (per rider)	\$12.75

Table 9: Fuel Consumption & other Assumptions on “Input-Start”

V. FUEL CONSUMPTION & OTHER ASSUMPTIONS

INSTRUCTION:

In this section, You are asked to give information about fuel consumption in order to estimate the fuel reduction, and other general assumptions. The Numbers shown in **pink cells** are the values from previously studies and used as default values to run the model. Please update it if you have different values in your case.

Assumptions about Fuel Consumption

Average Mileage of Consuming one Gallon of Fuel

25

Fuel Price per Gallon

\$1.10

Emission Rates during Acceleration (grams/gallon)

Nox

24.7

HC

9.5

CO

209.0

Emission Rates during Idling (grams/minute)

Nox

0.0

HC

0.15

CO

2.5

Cost of Air Pollution (\$/kg of pollutant)

Nox

1.275

HC

1.275

CO

0.0063

General Assumptions

Inflation Rate

3%

Annual Interest Rate

5%

Discount Rate

2%

Personal Year (PY)

Hours per Unit Personal Year

1,768

PY for Information Technology

0.11

PY for Program Costs (Toll Collection)

61.50

Unit Cost per Personal Year

\$65,000

Table 10: Annual Traffic Data on “*Input-Detail*”

Ia. TRAFFIC DATA - Annual

INSTRUCTION:

Change the numbers in blue cells only -

The data in Base Year and Year 20 you enter in sheet "Input-Start" are shown on **white cells**.

Based on these data, the model estimated automatically annual traffic volume and ETC Split in between by interpolation.

Please make reasonable adjustments in **blue cells**.

Estimated Annual Traffic Volume (ATV)

Year	Traffic Vol.	ETC Split
<i>Base</i>	19,016,173	0%
<i>1</i>	19,565,364	6%
<i>2</i>	20,114,556	15%
<i>3</i>	20,663,747	20%
<i>4</i>	21,212,938	25%
<i>5</i>	21,762,130	30%
<i>6</i>	22,311,321	35%
<i>7</i>	22,860,512	40%
<i>8</i>	23,409,704	45%
<i>9</i>	23,958,895	50%
<i>10</i>	24,508,087	55%
<i>11</i>	25,057,278	60%
<i>12</i>	25,606,469	65%
<i>13</i>	26,155,661	70%
<i>14</i>	26,704,852	75%
<i>15</i>	27,254,043	80%
<i>16</i>	27,803,235	85%
<i>17</i>	28,352,426	90%
<i>18</i>	28,901,617	95%
<i>19</i>	29,450,809	95%
<i>20</i>	30,000,000	95%

Table 11: Traffic Data in the Morning Peak on “Input-Detail”

Ib. TRAFFIC DATA - Morning Peak

INSTRUCTION:

Change the numbers in blue cells only -

The data in Base Year and Year 20 you enter in sheet "Input-Start" are shown on **white cells**.

Based on these data, the model estimated automatically peak hour traffic volume, duration of peak period, and D factor for each year.

- a) Peak Hour Volume (PHV): assume that ATV/PHV remains constant over the analysis period. Thus, peak hour volume in year i is estimated by estimated annual traffic volume times ATV/PHV.
- b) ATV/PHV - the ratio of annual traffic volume over peak hour volume.
- c) Duration of Peak Period (T) in the morning peak in Based Year, hrs/day.
- d) Direction Factor (D) during the morning peak in Base Year

Please make reasonable changes in **blue cells**.

Year	PHV	ATV/PHV	T (hr)	K
<i>Base</i>	8,200	2,319	2	0.3
<i>1</i>	8,437	2,319	2	0.3
<i>2</i>	8,674	2,319	2	0.3
<i>3</i>	8,910	2,319	2	0.3
<i>4</i>	9,147	2,319	2	0.3
<i>5</i>	9,384	2,319	2	0.3
<i>6</i>	9,621	2,319	2	0.3
<i>7</i>	9,858	2,319	2	0.3
<i>8</i>	10,095	2,319	2	0.3
<i>9</i>	10,331	2,319	2	0.3
<i>10</i>	10,568	2,319	2	0.3
<i>11</i>	10,805	2,319	2	0.3
<i>12</i>	11,042	2,319	2	0.3
<i>13</i>	11,279	2,319	2	0.3
<i>14</i>	11,515	2,319	2	0.3
<i>15</i>	11,752	2,319	2	0.3
<i>16</i>	11,989	2,319	2	0.3
<i>17</i>	12,226	2,319	2	0.3
<i>18</i>	12,463	2,319	2	0.3
<i>19</i>	12,700	2,319	2	0.3
<i>20</i>	12,936	2,319	2	0.3

Table 12: Costs Information for Baseline on “Input-Detail”

IIa. COSTS INFORMATION--Baseline (No ETC)

INSTRUCTION:

Change the numbers in blue cells only -

The data in Base Year and Year 20 you enter in sheet "Input-Start" are shown on **white cells after inflation effect**. That is, the number show on this table are no longer current value but nominal value in year i.

Based on these data, the model estimates continuing costs automatically.

For the knowledge of estimation, check the footnote right below the table.

Please make reasonable adjustments in **blue cells**.

Year	One-Time Costs	Continuing Costs:		
		Staff*	Operations/Materials	Other
<i>Base</i>	\$0	\$4,004,650	\$8,356	\$297,120
<i>1</i>	\$0	\$4,230,053	\$9,390	\$302,720
<i>2</i>	\$0	\$4,465,376	\$10,587	\$308,800
<i>3</i>	\$0	\$4,711,012	\$11,975	\$315,200
<i>4</i>	\$0	\$4,967,367	\$13,589	\$322,240
<i>5</i>	\$0	\$5,234,863	\$15,472	\$329,760
<i>6</i>	\$0	\$5,513,938	\$17,673	\$339,653
<i>7</i>	\$0	\$5,805,047	\$20,252	\$349,842
<i>8</i>	\$0	\$6,108,659	\$23,280	\$360,338
<i>9</i>	\$0	\$6,425,264	\$26,841	\$371,148
<i>10</i>	\$0	\$6,755,367	\$31,038	\$382,282
<i>11</i>	\$0	\$7,099,494	\$32,590	\$393,751
<i>12</i>	\$0	\$7,458,188	\$34,219	\$405,563
<i>13</i>	\$0	\$7,832,015	\$35,930	\$417,730
<i>14</i>	\$0	\$8,221,559	\$37,727	\$430,262
<i>15</i>	\$0	\$8,627,426	\$39,613	\$443,170
<i>16</i>	\$0	\$9,050,246	\$41,594	\$456,465
<i>17</i>	\$0	\$9,490,671	\$43,674	\$470,159
<i>18</i>	\$0	\$9,949,376	\$45,857	\$484,264
<i>19</i>	\$0	\$10,427,062	\$48,150	\$498,792
<i>20</i>	\$0	\$10,924,454	\$50,558	\$513,755

Table 13: Costs information for ETC Alternative on “Input-Detail”

IIb. COSTS INFORMATION--ETC Alternative

INSTRUCTION:

Change the numbers in blue cells only -

The data in Base Year and Year 20 you enter in sheet "Input-Start" are shown on **white cells after inflation effect**. That is, the number show on this table are no longer current value but nominal value.

Based on these data, the model estimated continuing cost automatically.

For the knowledge of estimation, check the footnote right below the table.

Please make reasonable adjustments in **blue cells**.

Year	Capital Investment	Continuing Costs:			
		One-Time Costs	Staff	Operations/Materials	Other
<i>Base</i>	\$3,009,340	\$0	\$4,004,650	\$8,356	\$301,760
<i>1</i>	\$0	\$231,886	\$3,947,124	\$9,390	\$325,760
<i>2</i>	\$0	\$244,904	\$3,811,406	\$3,600	\$122,080
<i>3</i>	\$0	\$257,922	\$3,811,185	\$3,780	\$133,760
<i>4</i>	\$0	\$270,940	\$3,753,045	\$3,969	\$133,920
<i>5</i>	\$0	\$283,958	\$3,720,527	\$4,167	\$134,080
<i>6</i>	\$0	\$296,976	\$3,670,541	\$4,376	\$138,102
<i>7</i>	\$0	\$309,994	\$3,554,679	\$4,595	\$142,245
<i>8</i>	\$0	\$323,012	\$3,464,120	\$4,824	\$146,513
<i>9</i>	\$0	\$336,029	\$3,351,664	\$5,066	\$150,908
<i>10</i>	\$0	\$349,047	\$3,164,295	\$5,319	\$153,495
<i>11</i>	\$0	\$362,065	\$3,001,522	\$5,585	\$158,100
<i>12</i>	\$0	\$375,083	\$2,757,126	\$5,864	\$162,843
<i>13</i>	\$0	\$388,101	\$2,536,586	\$6,157	\$167,728
<i>14</i>	\$0	\$401,119	\$2,227,120	\$6,465	\$172,760
<i>15</i>	\$0	\$414,137	\$1,880,965	\$6,788	\$177,943
<i>16</i>	\$0	\$427,155	\$1,495,724	\$7,128	\$183,281
<i>17</i>	\$0	\$440,173	\$1,068,871	\$7,484	\$188,779
<i>18</i>	\$0	\$453,190	\$597,755	\$7,858	\$194,443
<i>19</i>	\$0	\$466,208	\$624,601	\$8,251	\$200,276
<i>20</i>	\$0	\$479,226	\$652,519	\$8,664	\$206,284

IV. Expected Payment Split

The model assumed constant evolving rate from base year to year 20. We keep those assumptions; thus, make no change on this table. The results are shown on Table 14.

Table 14: Expected Payment Split on “Input-Detail”

IV. EXPECTED PAYMENT SPLIT

INSTRUCTION:

Change the numbers in blue cells only -
the numbers in white cells are those you entered in "Input-Start" Sheet about .
payment split among Manual Users as well as among ETC Users.

Payment Split

Year	Baseline		ETC Alternative			
	Manual Users		Manual Users		ETC Users	
	Cash	Ticket	Cash	Ticket	Cash & Checks	Credit Card
<i>Base</i>	83%	17%	83%	17%	64%	36%
<i>1</i>	83%	17%	84%	16%	64%	36%
<i>2</i>	83%	17%	85%	15%	64%	36%
<i>3</i>	83%	17%	86%	14%	64%	36%
<i>4</i>	83%	17%	86%	14%	64%	36%
<i>5</i>	83%	17%	87%	13%	64%	36%
<i>6</i>	83%	17%	88%	12%	64%	36%
<i>7</i>	83%	17%	89%	11%	64%	36%
<i>8</i>	83%	17%	90%	10%	64%	36%
<i>9</i>	83%	17%	91%	9%	64%	36%
<i>10</i>	83%	17%	92%	9%	64%	36%
<i>11</i>	83%	17%	92%	8%	64%	36%
<i>12</i>	83%	17%	93%	7%	64%	36%
<i>13</i>	83%	17%	94%	6%	64%	36%
<i>14</i>	83%	17%	95%	5%	64%	36%
<i>15</i>	83%	17%	96%	4%	64%	36%
<i>16</i>	83%	17%	97%	3%	64%	36%
<i>17</i>	83%	17%	97%	3%	64%	36%
<i>18</i>	83%	17%	98%	2%	64%	36%
<i>19</i>	83%	17%	99%	1%	64%	36%
<i>20</i>	83%	17%	100%	0%	64%	36%

Okay, let's check the result first. Suppose I am interested in Benefit/Cost ratio, I chose to go to "Results Summary" first.

Results Summary

You must be sort of tired after a lot of data input, in particular, what you want to know are relative simple: Net Present Value, B/C ratio, and maybe Internal Rate of Return. Overall net present value of project is presented as well as net present value in terms of toll agency, users, and community. The result of sample input is shown on Table 15.

The second table in this sheet is the result in case that analysis horizon is set as 10 years instead of 20 years.

Table 15: Analytical Results on "Results Summary"

☒ Analytical Results

☐ For 20 years

	NPV ¹ =	PVB ² =	NPC ³ =	B/C ⁴ =	IRR ⁵ =
Overall	\$102,375,456	\$105,384,796	\$3,009,340	35.02	48.44%
Toll Agency	\$55,966,011	\$58,975,351	\$3,009,340	19.60	
Users	\$46,345,558	\$46,345,558	\$0		
Community	\$63,887	\$63,887	\$0		

¹NPV: net present value, NPV = PVB - PVC

²PVB: present value of benefits

³PVC: present value of costs

⁴B/C: Total benefit over capital investment, B/C = PVB/PVC

⁵IRR: Internal Rate of Return

☐ For 10 years

	NPV ¹ =	PVB ² =	NPC ³ =	B/C ⁴ =	IRR ⁵ =
Overall	\$21,137,976	\$24,147,316	\$3,009,340	8.02	44.57%
Toll Agency	\$11,160,310	\$14,169,650	\$3,009,340	4.71	
Users	\$9,960,192	\$9,960,192	\$0		
Community	\$17,474	\$17,474	\$0		

Well, what can we say about these numbers? First of all, B/C ratios are greater than one in both cases. That means, capital investment in ETC system is valuable compared to do-nothing alternative. Second, toll agency is the only agent who has to pay for benefits. However, the amount of benefits as a result of initial capital investment is greater than of amount of capital expenditures. Positive NPV and B/C ratio can be used as justify the ETC project positively. Third, if the lifetime of ETC system is only ten years, the

monetary value of benefits gained with the same initial investment drops dramatically compared to the case with lifetime of 20 years. However, positive net present value and B/C ratio can still say that implementation of ETC project is better than doing nothing.

Now, you may want to know where benefits come from and the distribution of different kinds of benefits. Click on the sheet “Summary-Details” and solve your inquiries.

Results-Detail

Cost and benefit streams for each agents as well as whole society are tabulated in this sheet. Detailed results with sample inputs are shown on Table 16, Table 17, Table 18, and Table 19.

In a whole, the major benefits of ETC system is time saving for users and cost reduction for toll agency. Costs reduction for toll agency collects for about one half of the value of overall benefits. The main source of huge cost reduction is the reduced staff needed for toll collection. According to Table 3, staff costs is about 90% of total continuing costs in base year. As ETC market share expands over time, cost saving in staff is obviously significant.

At the same time, users save a lot of time by introducing ETC system. This implies that ETC system may solve to existing congestion suffering by commuters efficiently. The great amount of time saving also reflects that the accuracy of delay formula plays a crucial role in this analysis. Overestimated delay generates higher value of benefits that realizes.

Interests generates by ETC account is counted as positive operational revenues for toll agency but interest lost for users. Therefore, just like internal transfer within the system, the amounts of these two items cancel out; thus, not appear in overall benefit.

Cost reduction in purchasing fuel is another sort of benefit to users. It only accounts for two to three percent of the value of overall benefits. The value of benefit to community is relatively minor, compared to other giants. Usually, benefit cost analysis arise more questions from the discrepancies on the value of health and time. In addition to precision of the quantity of saving, equivalent monetary value of some qualitative benefits is controversial. You may want to change some assumptions on “Input-Start” to see how sensitive the results are.

The rest of sheets in this file show calculation process to estimate the value of a variety of benefits. Let us start with the major source of benefits: cost reduction of toll agency. In order to estimate the cost reduction, number of personal year needed must be known. While number of personal years required in year n is associated with the number of ETC lanes, and non-ETC traffic, we have to find the optimal number of ETC lanes first.

Table 16: Overall Benefits and Costs Streams

☒ Total Cost Streams

Year	Capital Investment	PVC (\$ in Base Year)
Base	\$3,009,340	\$3,009,340
1	\$0	\$0
2	\$0	\$0
3	\$0	\$0
4	\$0	\$0
5	\$0	\$0
6	\$0	\$0
7	\$0	\$0
8	\$0	\$0
9	\$0	\$0
10	\$0	\$0
11	\$0	\$0
12	\$0	\$0
13	\$0	\$0
14	\$0	\$0
15	\$0	\$0
16	\$0	\$0
17	\$0	\$0
18	\$0	\$0
19	\$0	\$0
20	\$0	\$0

☒ Total Benefit Streams

Year	ΔFuel Value	ΔTime Value	ΔEnv Value	Agency Cost Reduction	ΔOperational Revenue	ΔInterest Gain	Total Benefit	PVB (\$ in Base Year)
Base	\$0	\$0	\$0	-\$4,640	\$0	\$0	-\$4,640	(\$4,640)
1	\$8,178	\$97,151	\$299	\$2,506	\$20,215	-\$20,215	\$108,134	\$104,984
2	\$16,816	\$343,913	\$682	\$570,106	\$51,956	-\$51,956	\$931,516	\$878,043
3	\$25,912	\$530,378	\$1,044	\$789,943	\$71,166	-\$71,166	\$1,347,277	\$1,232,949
4	\$35,468	\$776,598	\$1,442	\$1,089,791	\$91,322	-\$91,322	\$1,903,299	\$1,691,057
5	\$45,483	\$1,058,901	\$1,864	\$1,374,852	\$112,423	-\$112,423	\$2,481,100	\$2,140,218
6	\$55,957	\$1,345,198	\$2,295	\$1,686,693	\$134,470	-\$134,470	\$3,090,142	\$2,587,946
7	\$66,890	\$1,667,969	\$2,748	\$2,075,858	\$157,463	-\$157,463	\$3,813,465	\$3,100,696
8	\$78,282	\$2,033,817	\$3,226	\$2,451,675	\$181,402	-\$181,402	\$4,567,000	\$3,605,232
9	\$90,133	\$2,408,483	\$3,713	\$2,861,875	\$206,286	-\$206,286	\$5,364,204	\$4,111,216
10	\$102,444	\$2,847,233	\$4,230	\$3,361,982	\$232,116	-\$232,116	\$6,315,889	\$4,699,615
11	\$115,213	\$3,303,461	\$4,760	\$3,845,866	\$258,892	-\$258,892	\$7,269,300	\$5,251,497
12	\$128,442	\$3,810,007	\$5,312	\$4,424,853	\$286,613	-\$286,613	\$8,368,614	\$5,869,578
13	\$142,130	\$4,346,931	\$5,880	\$4,993,987	\$315,280	-\$315,280	\$9,488,928	\$6,461,498
14	\$156,277	\$4,938,419	\$6,471	\$5,666,590	\$344,893	-\$344,893	\$10,767,756	\$7,118,756
15	\$170,883	\$5,570,695	\$7,078	\$6,390,990	\$375,452	-\$375,452	\$12,139,646	\$7,791,977
16	\$185,948	\$6,251,062	\$7,705	\$7,170,165	\$406,956	-\$406,956	\$13,614,880	\$8,484,343
17	\$201,472	\$6,984,377	\$8,350	\$8,007,244	\$439,406	-\$439,406	\$15,201,444	\$9,197,123
18	\$217,456	\$7,774,363	\$9,016	\$8,905,507	\$472,802	-\$472,802	\$16,906,341	\$9,930,694
19	\$233,898	\$8,586,595	\$9,690	\$9,320,849	\$481,786	-\$481,786	\$18,151,032	\$10,351,280
20	\$250,800	\$9,456,913	\$10,382	\$9,753,112	\$490,770	-\$490,770	\$19,471,208	\$10,780,736

☒ Total Net Benefit (\$ in base year)

Year	B-C	PVC	PVB	NPV
Base	-\$3,013,980	\$3,009,340	-\$4,640	(\$3,013,980)
1	\$108,134	\$0	\$104,984	\$104,984
2	\$931,516	\$0	\$878,043	\$878,043
3	\$1,347,277	\$0	\$1,232,949	\$1,232,949
4	\$1,903,299	\$0	\$1,691,057	\$1,691,057
5	\$2,481,100	\$0	\$2,140,218	\$2,140,218
6	\$3,090,142	\$0	\$2,587,946	\$2,587,946
7	\$3,813,465	\$0	\$3,100,696	\$3,100,696
8	\$4,567,000	\$0	\$3,605,232	\$3,605,232
9	\$5,364,204	\$0	\$4,111,216	\$4,111,216
10	\$6,315,889	\$0	\$4,699,615	\$4,699,615
11	\$7,269,300	\$0	\$5,251,497	\$5,251,497
12	\$8,368,614	\$0	\$5,869,578	\$5,869,578
13	\$9,488,928	\$0	\$6,461,498	\$6,461,498
14	\$10,767,756	\$0	\$7,118,756	\$7,118,756
15	\$12,139,646	\$0	\$7,791,977	\$7,791,977
16	\$13,614,880	\$0	\$8,484,343	\$8,484,343
17	\$15,201,444	\$0	\$9,197,123	\$9,197,123
18	\$16,906,341	\$0	\$9,930,694	\$9,930,694
19	\$18,151,032	\$0	\$10,351,280	\$10,351,280
20	\$19,471,208	\$0	\$10,780,736	\$10,780,736
Total		\$3,009,340	\$105,384,796	\$102,375,456

Table 17: Cost and Benefit Streams of Toll Agency

☞ Cost Streams of Toll Agency

Year	Capital Investment	PVC (\$ in Base Year)
Base	\$3,009,340	\$3,009,340
1	\$0	\$0
2	\$0	\$0
3	\$0	\$0
4	\$0	\$0
5	\$0	\$0
6	\$0	\$0
7	\$0	\$0
8	\$0	\$0
9	\$0	\$0
10	\$0	\$0
11	\$0	\$0
12	\$0	\$0
13	\$0	\$0
14	\$0	\$0
15	\$0	\$0
16	\$0	\$0
17	\$0	\$0
18	\$0	\$0
19	\$0	\$0
20	\$0	\$0

☞ Benefit Streams of Toll Agency

Year	Cost Reduction	ΔOperational Revenue*	Total Benefit	PVB (\$ in Base Year)
Base	-\$4,640	\$0	-\$4,640	(\$4,640)
1	\$2,506	\$20,215	\$22,720	\$22,059
2	\$570,106	\$51,956	\$622,062	\$586,353
3	\$789,943	\$71,166	\$861,109	\$788,037
4	\$1,089,791	\$91,322	\$1,181,113	\$1,049,403
5	\$1,374,852	\$112,423	\$1,487,275	\$1,282,937
6	\$1,686,693	\$134,470	\$1,821,163	\$1,525,196
7	\$2,075,858	\$157,463	\$2,233,322	\$1,815,895
8	\$2,451,675	\$181,402	\$2,633,077	\$2,078,575
9	\$2,861,875	\$206,286	\$3,068,161	\$2,351,490
10	\$3,361,982	\$232,116	\$3,594,098	\$2,674,347
11	\$3,845,866	\$258,892	\$4,104,758	\$2,965,364
12	\$4,424,853	\$286,613	\$4,711,466	\$3,304,528
13	\$4,993,987	\$315,280	\$5,309,267	\$3,615,353
14	\$5,666,590	\$344,893	\$6,011,484	\$3,974,299
15	\$6,390,990	\$375,452	\$6,766,441	\$4,343,121
16	\$7,170,165	\$406,956	\$7,577,121	\$4,721,811
17	\$8,007,244	\$439,406	\$8,446,650	\$5,110,362
18	\$8,905,507	\$472,802	\$9,378,308	\$5,508,768
19	\$9,320,849	\$481,786	\$9,802,635	\$5,590,306
20	\$9,753,112	\$490,770	\$10,243,882	\$5,671,789

*ΔOperational Benefit = the amount of DInterest Gain of users

☞ Net Benefit of Toll Agency (\$ in base year)

Year	PVC	PVB	NPV
Base	\$3,009,340	-\$4,640	(\$3,013,980)
1	\$0	\$22,059	\$22,059
2	\$0	\$586,353	\$586,353
3	\$0	\$788,037	\$788,037
4	\$0	\$1,049,403	\$1,049,403
5	\$0	\$1,282,937	\$1,282,937
6	\$0	\$1,525,196	\$1,525,196
7	\$0	\$1,815,895	\$1,815,895
8	\$0	\$2,078,575	\$2,078,575
9	\$0	\$2,351,490	\$2,351,490
10	\$0	\$2,674,347	\$2,674,347
11	\$0	\$2,965,364	\$2,965,364
12	\$0	\$3,304,528	\$3,304,528
13	\$0	\$3,615,353	\$3,615,353
14	\$0	\$3,974,299	\$3,974,299
15	\$0	\$4,343,121	\$4,343,121
16	\$0	\$4,721,811	\$4,721,811
17	\$0	\$5,110,362	\$5,110,362
18	\$0	\$5,508,768	\$5,508,768
19	\$0	\$5,590,306	\$5,590,306
20	\$0	\$5,671,789	\$5,671,789
Total	\$3,009,340	\$58,975,351	\$55,966,011

Table 18: Cost and Benefit Streams of Users

☞ Cost Streams of Users

☞ Benefit Streams of Users

☞ Net Benefit of Users (\$ in base year)

Year	Capital Investment	PVC (\$ in Base Year)	Year	ΔFuel Value	ΔTime Value	ΔInterest Gain	ΔTotal	PVB (\$ in Base Year)	Year	PVC	PVB	NPV
Base	\$0	\$0	Base	\$0	\$0	\$0	\$0	\$0	Base	\$0	\$0	\$0
1	\$0	\$0	1	\$8,178	\$97,151	-\$20,215	\$85,114	\$82,635	1	\$0	\$82,635	\$82,635
2	\$0	\$0	2	\$16,816	\$343,913	-\$51,956	\$308,772	\$291,048	2	\$0	\$291,048	\$291,048
3	\$0	\$0	3	\$25,912	\$530,378	-\$71,166	\$485,124	\$443,957	3	\$0	\$443,957	\$443,957
4	\$0	\$0	4	\$35,468	\$776,598	-\$91,322	\$720,744	\$640,372	4	\$0	\$640,372	\$640,372
5	\$0	\$0	5	\$45,483	\$1,058,901	-\$112,423	\$991,961	\$855,674	5	\$0	\$855,674	\$855,674
6	\$0	\$0	6	\$55,957	\$1,345,198	-\$134,470	\$1,266,684	\$1,060,828	6	\$0	\$1,060,828	\$1,060,828
7	\$0	\$0	7	\$66,890	\$1,667,969	-\$157,463	\$1,577,395	\$1,282,567	7	\$0	\$1,282,567	\$1,282,567
8	\$0	\$0	8	\$78,282	\$2,033,817	-\$181,402	\$1,930,697	\$1,524,110	8	\$0	\$1,524,110	\$1,524,110
9	\$0	\$0	9	\$90,133	\$2,408,483	-\$206,286	\$2,292,330	\$1,756,880	9	\$0	\$1,756,880	\$1,756,880
10	\$0	\$0	10	\$102,444	\$2,847,233	-\$232,116	\$2,717,561	\$2,022,120	10	\$0	\$2,022,120	\$2,022,120
11	\$0	\$0	11	\$115,213	\$3,303,461	-\$258,892	\$3,159,782	\$2,282,694	11	\$0	\$2,282,694	\$2,282,694
12	\$0	\$0	12	\$128,442	\$3,810,007	-\$286,613	\$3,651,836	\$2,561,324	12	\$0	\$2,561,324	\$2,561,324
13	\$0	\$0	13	\$142,130	\$4,346,931	-\$315,280	\$4,173,781	\$2,842,141	13	\$0	\$2,842,141	\$2,842,141
14	\$0	\$0	14	\$156,277	\$4,938,419	-\$344,893	\$4,749,802	\$3,140,179	14	\$0	\$3,140,179	\$3,140,179
15	\$0	\$0	15	\$170,883	\$5,570,695	-\$375,452	\$5,366,126	\$3,444,312	15	\$0	\$3,444,312	\$3,444,312
16	\$0	\$0	16	\$185,948	\$6,251,062	-\$406,956	\$6,030,054	\$3,757,730	16	\$0	\$3,757,730	\$3,757,730
17	\$0	\$0	17	\$201,472	\$6,984,377	-\$439,406	\$6,746,443	\$4,081,709	17	\$0	\$4,081,709	\$4,081,709
18	\$0	\$0	18	\$217,456	\$7,774,363	-\$472,802	\$7,519,017	\$4,416,630	18	\$0	\$4,416,630	\$4,416,630
19	\$0	\$0	19	\$233,898	\$8,586,595	-\$481,786	\$8,338,707	\$4,755,448	19	\$0	\$4,755,448	\$4,755,448
20	\$0	\$0	20	\$250,800	\$9,456,913	-\$490,770	\$9,216,943	\$5,103,198	20	\$0	\$5,103,198	\$5,103,198
Total									Total	\$0	\$46,345,558	\$46,345,558

Table 19: Cost and Benefit Streams of Community

☞ Cost Streams of Community

Year	Capital Investment	PVC (\$ in Base Year)
Base	\$0	\$0
1	\$0	\$0
2	\$0	\$0
3	\$0	\$0
4	\$0	\$0
5	\$0	\$0
6	\$0	\$0
7	\$0	\$0
8	\$0	\$0
9	\$0	\$0
10	\$0	\$0
11	\$0	\$0
12	\$0	\$0
13	\$0	\$0
14	\$0	\$0
15	\$0	\$0
16	\$0	\$0
17	\$0	\$0
18	\$0	\$0
19	\$0	\$0
20	\$0	\$0

☞ Benefit Streams of Community

Year	ΔEnv. Value**	ΔTotal	PVB
Base	\$0	\$0	\$0
1	\$299	\$299	\$290
2	\$682	\$682	\$643
3	\$1,044	\$1,044	\$955
4	\$1,442	\$1,442	\$1,281
5	\$1,864	\$1,864	\$1,608
6	\$2,295	\$2,295	\$1,922
7	\$2,748	\$2,748	\$2,235
8	\$3,226	\$3,226	\$2,546
9	\$3,713	\$3,713	\$2,845
10	\$4,230	\$4,230	\$3,148
11	\$4,760	\$4,760	\$3,439
12	\$5,312	\$5,312	\$3,726
13	\$5,880	\$5,880	\$4,004
14	\$6,471	\$6,471	\$4,278
15	\$7,078	\$7,078	\$4,543
16	\$7,705	\$7,705	\$4,801
17	\$8,350	\$8,350	\$5,052
18	\$9,016	\$9,016	\$5,296
19	\$9,690	\$9,690	\$5,526
20	\$10,382	\$10,382	\$5,748

** Environmental value is estimated by the value of emission reduction.

☞ Net Benefit of Community (\$ in base year)

Year	PVC	PVB	NPV
Base	\$0	\$0	\$0
1	\$0	\$290	\$290
2	\$0	\$643	\$643
3	\$0	\$955	\$955
4	\$0	\$1,281	\$1,281
5	\$0	\$1,608	\$1,608
6	\$0	\$1,922	\$1,922
7	\$0	\$2,235	\$2,235
8	\$0	\$2,546	\$2,546
9	\$0	\$2,845	\$2,845
10	\$0	\$3,148	\$3,148
11	\$0	\$3,439	\$3,439
12	\$0	\$3,726	\$3,726
13	\$0	\$4,004	\$4,004
14	\$0	\$4,278	\$4,278
15	\$0	\$4,543	\$4,543
16	\$0	\$4,801	\$4,801
17	\$0	\$5,052	\$5,052
18	\$0	\$5,296	\$5,296
19	\$0	\$5,526	\$5,526
20	\$0	\$5,748	\$5,748
Total	\$0	\$63,887	\$63,887

Table 20: Average Random Delay per Manual User during the Evening Peak (sec)

Year	ETC (%)	Total Arrival Rate	Arrival Rate for Manual Lanes	Number of ETC Lanes																					Mean Transaction Time (sec)			
				λ/sc for no ETC	Manual Only	λ/sc under ETC	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18		19	20	21
Base	0%	5,740	5,740	10.5	3.8	10.5	11	187	599	1,121	1,794	2,693	3,951	5,838	8,983	15,274	34,148	-	-	-	-	-	-	-	-	-	-	6.58
1	6%	5,906	5,551	10.1	3.0	10.2	7	89	475	981	1,634	2,506	3,726	5,557	8,609	14,713	33,026	-	-	-	-	-	-	-	-	-	-	6.60
2	15%	6,072	5,161	9.4	2.0	9.5	4	12	208	674	1,282	2,095	3,233	4,941	7,787	13,480	30,559	-	-	-	-	-	-	-	-	-	-	6.62
3	20%	6,237	4,990	9.1	1.7	9.2	3	7	106	547	1,136	1,924	3,028	4,685	7,445	12,967	29,534	-	-	-	-	-	-	-	-	-	-	6.64
4	25%	6,403	4,802	8.8	1.5	8.9	3	5	33	406	974	1,734	2,800	4,399	7,064	12,396	28,391	-	-	-	-	-	-	-	-	-	-	6.66
5	30%	6,569	4,598	8.4	1.3	8.5	2	4	13	253	794	1,524	2,548	4,084	6,644	11,765	27,129	-	-	-	-	-	-	-	-	-	-	6.68
6	35%	6,735	4,378	8.0	1.1	8.2	2	3	7	99	598	1,294	2,272	3,738	6,184	11,075	25,748	-	-	-	-	-	-	-	-	-	-	6.70
7	40%	6,900	4,140	7.6	0.9	7.7	1	2	5	21	387	1,045	1,971	3,363	5,683	10,324	24,246	-	-	-	-	-	-	-	-	-	-	6.72
8	45%	7,066	3,886	7.1	0.8	7.3	1	2	3	8	166	775	1,647	2,957	5,142	9,512	22,622	-	-	-	-	-	-	-	-	-	-	6.75
9	50%	7,232	3,616	6.6	0.7	6.8	1	1	2	5	26	487	1,298	2,521	4,560	8,638	20,875	-	-	-	-	-	-	-	-	-	-	6.77
10	55%	7,398	3,329	6.1	0.6	6.3	1	1	2	3	8	189	926	2,054	3,936	7,703	19,003	-	-	-	-	-	-	-	-	-	-	6.79
11	60%	7,563	3,025	5.5	0.5	5.7	1	1	1	2	4	21	530	1,555	3,271	6,705	17,007	-	-	-	-	-	-	-	-	-	-	6.81
12	65%	7,729	2,705	4.9	0.4	5.1	1	1	1	2	3	7	133	1,027	2,564	5,644	14,885	-	-	-	-	-	-	-	-	-	-	6.83
13	70%	7,895	2,369	4.3	0.3	4.5	0	1	1	1	2	3	12	472	1,816	4,520	12,636	-	-	-	-	-	-	-	-	-	-	6.85
14	75%	8,061	2,015	3.7	0.2	3.8	0	0	1	1	1	2	5	35	1,027	3,332	10,259	-	-	-	-	-	-	-	-	-	-	6.87
15	80%	8,227	1,645	3.0	0.2	3.2	0	0	0	1	1	1	2	6	220	2,081	7,752	-	-	-	-	-	-	-	-	-	-	6.89
16	85%	8,392	1,259	2.3	0.1	2.4	0	0	0	0	1	1	1	3	9	772	5,117	-	-	-	-	-	-	-	-	-	-	6.92
17	90%	8,558	856	1.6	0.1	1.6	0	0	0	0	0	0	1	1	3	16	2,354	-	-	-	-	-	-	-	-	-	-	6.94
18	95%	8,724	436	0.8	0.0	0.8	0	0	0	0	0	0	0	0	1	3	35	-	-	-	-	-	-	-	-	-	-	6.96
19	95%	8,890	444	0.8	0.0	0.9	0	0	0	0	0	0	0	0	1	3	40	-	-	-	-	-	-	-	-	-	-	6.98
20	95%	9,055	453	0.8	0.0	0.9	0	0	0	0	0	0	0	0	1	3	47	-	-	-	-	-	-	-	-	-	-	7.00

[illegible]

Table 22: Estimated Mean Service Rate per hour

Year	Payment Split (Baseline)-Manual			Payment Split (ETC Alternative)-Manual			Capacity-ETC lane
	Cash/Chect	Ticket	Mean Service Rate	Cash/Check	Ticket	Mean Service Rate	
<i>Base</i>	83%	17%	548	83%	17%	548	1500
<i>1</i>	83%	17%	548	84%	16%	546	1500
<i>2</i>	83%	17%	548	85%	15%	544	1500
<i>3</i>	83%	17%	548	86%	14%	542	1500
<i>4</i>	83%	17%	548	86%	14%	541	1500
<i>5</i>	83%	17%	548	87%	13%	539	1500
<i>6</i>	83%	17%	548	88%	12%	537	1500
<i>7</i>	83%	17%	548	89%	11%	535	1500
<i>8</i>	83%	17%	548	90%	10%	534	1500
<i>9</i>	83%	17%	548	91%	9%	532	1500
<i>10</i>	83%	17%	548	92%	9%	530	1500
<i>11</i>	83%	17%	548	92%	8%	529	1500
<i>12</i>	83%	17%	548	93%	7%	527	1500
<i>13</i>	83%	17%	548	94%	6%	525	1500
<i>14</i>	83%	17%	548	95%	5%	524	1500
<i>15</i>	83%	17%	548	96%	4%	522	1500
<i>16</i>	83%	17%	548	97%	3%	521	1500
<i>17</i>	83%	17%	548	97%	3%	519	1500
<i>18</i>	83%	17%	548	98%	2%	517	1500
<i>19</i>	83%	17%	548	99%	1%	516	1500
<i>20</i>	83%	17%	548	100%	0%	514	1500

Table 23: Total Random and Stop Delay in the System per day during the evening peak (hr)

Year	Traffic Vol.	ETC (%)	PHV	ATV/PHV	T (hr)	K	Total Arrival Rate	Number of ETC Lanes											Optimal # of ETC Ln*	Delay per Day** (hr)	Suggested # of ETC Ln***	Delay per Day** (hr)	
								Manual Only	1	2	3	4	5	6	7	8	9	10					11
Base	19,016,173	0%	8,200	2,319	2	0.7	5,740	33															
1	19,565,364	6%	8,437	2,319	2	0.7	5,906	31	43	295	1,485	3,046	5,061	7,748	11,513	17,160	26,573	45,398	101,876	1	43	1	43
2	20,114,556	15%	8,674	2,319	2	0.7	6,072	29	32	52	616	1,952	3,696	6,025	9,289	14,185	22,346	38,668	87,635	1	32	1	32
3	20,663,747	20%	8,910	2,319	2	0.7	6,237	29	35	40	313	1,535	3,168	5,353	8,413	13,005	20,658	35,966	81,890	1	35	1	35
4	21,212,938	25%	9,147	2,319	2	0.7	6,403	29	269	33	107	1,101	2,615	4,644	7,487	11,754	18,866	33,090	75,764	2	33	2	33
5	21,762,130	30%	9,384	2,319	2	0.7	6,569	29	1,270	29	50	663	2,046	3,910	6,525	10,449	16,990	30,072	69,320	2	29	2	29
6	22,311,321	35%	9,621	2,319	2	0.7	6,735	29	2,723	29	34	257	1,471	3,164	5,541	9,108	15,055	26,949	62,634	2	29	2	29
7	22,860,512	40%	9,858	2,319	2	0.7	6,900	29	4,665	41	28	64	905	2,419	4,550	7,751	13,087	23,761	55,784	3	28	3	28
8	23,409,704	45%	10,095	2,319	2	0.7	7,066	29	7,147	434	25	34	375	1,689	3,571	6,400	11,116	20,551	48,857	3	25	3	25
9	23,958,895	50%	10,331	2,319	2	0.7	7,232	29	10,226	1,515	25	25	66	993	2,622	5,078	9,173	17,367	41,948	3	25	3	25
10	24,508,087	55%	10,568	2,319	2	0.7	7,398	29	13,958	2,924	32	21	29	362	1,725	3,811	7,292	14,258	35,158	4	21	4	21
11	25,057,278	60%	10,805	2,319	2	0.7	7,563	30	18,405	4,675	193	20	21	47	904	2,626	5,510	11,281	28,597	4	20	4	20
12	25,606,469	65%	11,042	2,319	2	0.7	7,729	30	23,627	6,799	1,203	21	17	22	211	1,554	3,865	8,493	22,382	5	17	5	17
13	26,155,661	70%	11,279	2,319	2	0.7	7,895	30	29,690	9,326	2,545	31	15	15	26	630	2,399	5,957	16,636	5	15	5	15
14	26,704,852	75%	11,515	2,319	2	0.7	8,061	31	36,660	12,291	4,172	218	16	13	14	47	1,158	3,738	11,493	6	13	6	13
15	27,254,043	80%	11,752	2,319	2	0.7	8,227	31	44,606	15,728	6,104	1,306	20	11	11	13	209	1,909	7,093	7	11	7	11
16	27,803,235	85%	11,989	2,319	2	0.7	8,392	31	53,599	19,671	8,363	2,715	40	11	9	8	13	546	3,584	8	8	8	8
17	28,352,426	90%	12,226	2,319	2	0.7	8,558	32	63,712	24,158	10,974	4,386	486	14	8	6	6	12	1,123	9	6	9	6
18	28,901,617	95%	12,463	2,319	2	0.7	8,724	32	75,021	29,226	13,962	6,332	1,766	23	8	5	4	4	11	10	4	10	4
19	29,450,809	95%	12,700	2,319	2	0.7	8,890	33	78,220	30,668	14,818	6,895	2,150	30	8	5	4	4	13	10	4	10	4
20	30,000,000	95%	12,936	2,319	2	0.7	9,055	33	81,485	32,143	15,696	7,475	2,549	41	9	6	4	4	15	10	4	10	4

For each year, the allocation of lanes that has minimal total random and stop delay is called the optimal number of ETC lanes given total number of lanes. Before ETC traffic achieves a certain level, the optimal allocation of ETC lanes may cause average delay suffered by ETC users greater than manual users. Since marginal cost (time) of one more ETC user is far less than additional manual user. However, switch to ETC lanes is the way to ease congestion in the existing system. Therefore, we check the average delay for manual users and ETC users under optimal number of lanes, and come up with suggested number of lanes.

Table 24 shows the estimated total delay for a year. Suppose that peak hour only makes sense in weekdays, the total delay in the system per year is equal to total delay per day times the number of weekdays in a year (260). Because of unbalanced flow in the morning and in the evening, the suggested number of ETC lanes given total number of lanes are decided by one that peak traffic level is higher.

The queue is assumed not to occupy physical space in this analysis. Therefore, the amount of random and stop delay is equal to the idling time of vehicles.

Knowing the number of ETC lanes in year n , we are able to estimate the number of personal year needed now.

PY

The main benefit of ETC system from toll agency's point of view is costs reduction in the staff for toll collection. In estimating the units of PY needed in year n , we have several assumptions. First, manual lanes are all open during the peak period, while some of them are close at the rest of the day. Second, the number of PYs is well-balanced with current traffic level. Thus, a linear relationship is assumed between the units of PY needed during the off-peak and the off-peak traffic volume.

The estimation was shown on Table 25 and Table 26. First of all, the number of personal years during the peak hour is easy to estimate: sum up the peak period per day, multiply by the number of weekdays in a year, and divide by the hours per unit PY (1,768 in this case). Again, from the previous section, the number of suggestion ETC lanes decides the total number of manual lanes in a specific year.

To estimate the units of PY for off-peak operation, off-peak traffic volume has to be known first. Basically, we estimated the off-peak traffic volume in a year by subtracting annual peak traffic volume from annual traffic volume. We know the duration of peak period from the input sheet and the estimates for peak hour volume from the previous section. Daily peak traffic volume is estimated by the PHV in the morning times daily peak period in the morning and so for the evening. Multiplying daily peak traffic volume by the number of weekdays in a year gives us annual peak traffic volume. Scaling up the units of PY for the off-peak operation in base year to corresponding off-peak traffic volume gives us the units of PY for off-peak in that year.

Table 24: Total Random and Stop Delay per year (hr)

Year	Traffic Vol.	ETC (%)	Peak Arrival Rate in the evening	Delay per Day (Baseline)			Delay Per Year* (hr)	Delay per Day (ETC)			Delay Per Year* (hr)
				Morning	Evening	Total		Morning	Evening	Total	
<i>Base</i>	19,016,173	0%	5740								
<i>1</i>	19,565,364	6%	5,906	10	31	41	10,667	9	43	52	13,489
<i>2</i>	20,114,556	15%	6,072	10	29	39	10,113	9	32	40	10,479
<i>3</i>	20,663,747	20%	6,237	10	29	39	10,145	8	35	43	11,300
<i>4</i>	21,212,938	25%	6,403	11	29	39	10,194	8	33	41	10,705
<i>5</i>	21,762,130	30%	6,569	11	29	39	10,262	8	29	37	9,723
<i>6</i>	22,311,321	35%	6,735	11	29	40	10,345	8	29	37	9,565
<i>7</i>	22,860,512	40%	6,900	11	29	40	10,442	7	28	35	9,112
<i>8</i>	23,409,704	45%	7,066	12	29	41	10,552	7	25	32	8,231
<i>9</i>	23,958,895	50%	7,232	12	29	41	10,674	6	25	31	8,113
<i>10</i>	24,508,087	55%	7,398	12	29	42	10,805	6	21	27	7,035
<i>11</i>	25,057,278	60%	7,563	13	30	42	10,945	5	20	25	6,547
<i>12</i>	25,606,469	65%	7,729	13	30	43	11,093	5	17	22	5,699
<i>13</i>	26,155,661	70%	7,895	13	30	43	11,248	4	15	20	5,153
<i>14</i>	26,704,852	75%	8,061	13	31	44	11,409	4	13	16	4,282
<i>15</i>	27,254,043	80%	8,227	14	31	45	11,575	3	11	14	3,548
<i>16</i>	27,803,235	85%	8,392	14	31	45	11,746	2	8	11	2,819
<i>17</i>	28,352,426	90%	8,558	14	32	46	11,921	2	6	8	2,041
<i>18</i>	28,901,617	95%	8,724	14	32	47	12,100	1	4	5	1,198
<i>19</i>	29,450,809	95%	8,890	15	33	47	12,343	1	4	5	1,247
<i>20</i>	30,000,000	95%	9,055	15	33	48	12,587	1	4	5	1,297

Table 25: Estimation of the units of Personal Year-Part One

Year	Annual Traffic Vol.	AADT	Baseline			ETC (%)	Suggested Number of ETC Ln	ETC Alternative			PY Reduction
			PHV per year	Off-Peak Traffic per Year	PY			PHV (Manual) per year	Off-Peak Traffic (Manual) per Year	PY	
<i>Base</i>	19,016,173	52,099	4,264,000	14,752,173	61.50	0%		4,264,000	14,752,173	61.50	0
<i>1</i>	19,565,364	53,604	4,387,145	15,178,219	63.07	6%	1	4,123,917	14,267,526	59.12	4
<i>2</i>	20,114,556	55,108	4,510,291	15,604,265	64.64	15%	1	3,833,747	13,263,625	55.42	9
<i>3</i>	20,663,747	56,613	4,633,436	16,030,311	66.22	20%	1	3,706,749	12,824,249	53.80	12
<i>4</i>	21,212,938	58,118	4,756,581	16,456,357	67.79	25%	2	3,567,436	12,342,268	51.43	16
<i>5</i>	21,762,130	59,622	4,879,726	16,882,403	69.36	30%	2	3,415,808	11,817,682	49.49	20
<i>6</i>	22,311,321	61,127	5,002,872	17,308,449	70.93	35%	2	3,251,867	11,250,492	47.40	24
<i>7</i>	22,860,512	62,632	5,126,017	17,734,496	72.51	40%	3	3,075,610	10,640,697	44.56	28
<i>8</i>	23,409,704	64,136	5,249,162	18,160,542	74.08	45%	3	2,887,039	9,988,298	42.15	32
<i>9</i>	23,958,895	65,641	5,372,308	18,586,588	75.65	50%	3	2,686,154	9,293,294	39.59	36
<i>10</i>	24,508,087	67,145	5,495,453	19,012,634	77.22	55%	4	2,472,954	8,555,685	36.28	41
<i>11</i>	25,057,278	68,650	5,618,598	19,438,680	78.80	60%	4	2,247,439	7,775,472	33.40	45
<i>12</i>	25,606,469	70,155	5,741,743	19,864,726	80.37	65%	5	2,009,610	6,952,654	29.78	51
<i>13</i>	26,155,661	71,659	5,864,889	20,290,772	81.94	70%	5	1,759,467	6,087,232	26.58	55
<i>14</i>	26,704,852	73,164	5,988,034	20,716,818	83.51	75%	6	1,497,008	5,179,205	22.64	61
<i>15</i>	27,254,043	74,669	6,111,179	21,142,864	85.08	80%	7	1,222,236	4,228,573	18.55	67
<i>16</i>	27,803,235	76,173	6,234,324	21,568,910	86.66	85%	8	935,149	3,235,337	14.29	72
<i>17</i>	28,352,426	77,678	6,357,470	21,994,956	88.23	90%	9	635,747	2,199,496	9.88	78
<i>18</i>	28,901,617	79,183	6,480,615	22,421,002	89.80	95%	10	324,031	1,121,050	5.31	84
<i>19</i>	29,450,809	80,687	6,603,760	22,847,048	91.37	95%	10	330,188	1,142,352	5.39	86
<i>20</i>	30,000,000	82,192	6,726,906	23,273,094	92.95	95%	10	336,345	1,163,655	5.47	87

Table 26: Estimation of the Units of Personal Year Needed – Part Two

Year	Annual Traffic Vol.	Baseline			ETC (%)	ETC Alternative		
		PY for Peak Hour ¹	PY for Off-Peak per Year ²	Total		PY for Peak Hour ¹	PY for Off-Peak per Year ²	Total
<i>Base</i>	19,016,173	7.06	54.44	61.50	0%	7.06	54.44	61.50
<i>1</i>	19,565,364	7.06	56.01	63.07	6%	6.47	52.65	59.12
<i>2</i>	20,114,556	7.06	57.59	64.64	15%	6.47	48.95	55.42
<i>3</i>	20,663,747	7.06	59.16	66.22	20%	6.47	47.33	53.80
<i>4</i>	21,212,938	7.06	60.73	67.79	25%	5.88	45.55	51.43
<i>5</i>	21,762,130	7.06	62.30	69.36	30%	5.88	43.61	49.49
<i>6</i>	22,311,321	7.06	63.87	70.93	35%	5.88	41.52	47.40
<i>7</i>	22,860,512	7.06	65.45	72.51	40%	5.29	39.27	44.56
<i>8</i>	23,409,704	7.06	67.02	74.08	45%	5.29	36.86	42.15
<i>9</i>	23,958,895	7.06	68.59	75.65	50%	5.29	34.30	39.59
<i>10</i>	24,508,087	7.06	70.16	77.22	55%	4.71	31.57	36.28
<i>11</i>	25,057,278	7.06	71.74	78.80	60%	4.71	28.69	33.40
<i>12</i>	25,606,469	7.06	73.31	80.37	65%	4.12	25.66	29.78
<i>13</i>	26,155,661	7.06	74.88	81.94	70%	4.12	22.46	26.58
<i>14</i>	26,704,852	7.06	76.45	83.51	75%	3.53	19.11	22.64
<i>15</i>	27,254,043	7.06	78.03	85.08	80%	2.94	15.61	18.55
<i>16</i>	27,803,235	7.06	79.60	86.66	85%	2.35	11.94	14.29
<i>17</i>	28,352,426	7.06	81.17	88.23	90%	1.76	8.12	9.88
<i>18</i>	28,901,617	7.06	82.74	89.80	95%	1.18	4.14	5.31
<i>19</i>	29,450,809	7.06	84.31	91.37	95%	1.18	4.22	5.39
<i>20</i>	30,000,000	7.06	85.89	92.95	95%	1.18	4.29	5.47

It is time now to check how much the toll agency has to spend under different scenarios.

Agency Cost Reduction

The result is shown on Table 27. All costs information other than staff from input sheets. As for staff costs, multiplying the units of PY estimated in previous section by the costs of unit PY (\$65,000 in this case), and taking into account the inflation effect (3% per year) gives the answer. The cost of Staff accounts for 90% of total costs in base year. By introducing ETC system, this cost drops to less than one tenth corresponding to estimated expenditures for staff under baseline.

Certain amount of money transfer from the users to the toll agency as a result of required minimum balance in ETC accounts. From the users' point of view, it is interest loss; whereas from the viewpoint of toll agency, it is operational revenue. The amount of interest transfers between these two agents is estimated in the next section.

Interest Gain

The amount of interest loss from ETC users is the amount of money users put in their account times the annual interest rate (6% in this case). In the case of Carquinez Bridge, ETC accounts can be established in two ways: Cash & Check, or Credit Card. The ETC accounts established with cash or checks need additional \$30 as deposit for transponders, whereas accounts with credit cards don't. According to ETC account fact on Table 7, average balance other than deposits in both types of accounts is \$20. Question arose from how to relate the number of accounts to information we have, say annual traffic level. Based on Table 7, again, we found that average usage per ETC account per year is 160, and average tags per account is 1.35. The number of tags is needed for estimating interest loss from deposits. The number of tags greater than 1 told the fact several vehicles may share a ETC account, such as family members with more than one car.

The result is shown on Table 28. The number of ETC transaction is estimated by annual traffic volume and expected ETC split. The Payment Split among ETC users is shown on Table 14 or "*Input-Detail*" in the file. Combining above information with average tags and average usage per account, the number of accounts for each type of payment is estimated. The amount of money, including deposits, and the number of accounts in each type of payments gives the amount of total balance in a year. Multiplying this number by the annual interest rate is the interest loss in a year shown on the last column on the table.

Moving away from the standing point of the toll agency, we will see how to get the value of time saving for users in the next section.

Table 27: Agency Costs Estimation

Year	Baseline					ETC Alternative					Cost Saving
	One-Time	Staff	Operations/Materials	Others	Total	One-Time	Staff	Operations/Materials	Others	Total	
<i>Base</i>	\$0	\$4,004,650	\$8,356	\$297,120	\$4,310,126	\$0	\$4,004,650	\$8,356	\$301,760	\$4,314,766	(\$4,640)
<i>1</i>	\$0	\$4,230,053	\$9,390	\$302,720	\$4,542,164	\$238,843	\$3,965,665	\$9,390	\$325,760	\$4,539,658	\$2,506
<i>2</i>	\$0	\$4,465,376	\$10,587	\$308,800	\$4,784,763	\$259,819	\$3,829,159	\$3,600	\$122,080	\$4,214,658	\$570,106
<i>3</i>	\$0	\$4,711,012	\$11,975	\$315,200	\$5,038,187	\$281,838	\$3,828,865	\$3,780	\$133,760	\$4,248,244	\$789,943
<i>4</i>	\$0	\$4,967,367	\$13,589	\$322,240	\$5,303,196	\$304,945	\$3,770,571	\$3,969	\$133,920	\$4,213,405	\$1,089,791
<i>5</i>	\$0	\$5,234,863	\$15,472	\$329,760	\$5,580,095	\$329,185	\$3,737,811	\$4,167	\$134,080	\$4,205,243	\$1,374,852
<i>6</i>	\$0	\$5,513,938	\$17,673	\$339,653	\$5,871,265	\$354,605	\$3,687,489	\$4,376	\$138,102	\$4,184,572	\$1,686,693
<i>7</i>	\$0	\$5,805,047	\$20,252	\$349,842	\$6,175,141	\$381,253	\$3,571,190	\$4,595	\$142,245	\$4,099,283	\$2,075,858
<i>8</i>	\$0	\$6,108,659	\$23,280	\$360,338	\$6,492,277	\$409,181	\$3,480,083	\$4,824	\$146,513	\$4,040,602	\$2,451,675
<i>9</i>	\$0	\$6,425,264	\$26,841	\$371,148	\$6,823,253	\$438,442	\$3,366,962	\$5,066	\$150,908	\$3,961,378	\$2,861,875
<i>10</i>	\$0	\$6,755,367	\$31,038	\$382,282	\$7,168,688	\$469,090	\$3,178,801	\$5,319	\$153,495	\$3,806,705	\$3,361,982
<i>11</i>	\$0	\$7,099,494	\$32,590	\$393,751	\$7,525,835	\$501,183	\$3,015,101	\$5,585	\$158,100	\$3,679,968	\$3,845,866
<i>12</i>	\$0	\$7,458,188	\$34,219	\$405,563	\$7,897,971	\$534,779	\$2,769,632	\$5,864	\$162,843	\$3,473,118	\$4,424,853
<i>13</i>	\$0	\$7,832,015	\$35,930	\$417,730	\$8,285,675	\$569,939	\$2,547,864	\$6,157	\$167,728	\$3,291,689	\$4,993,987
<i>14</i>	\$0	\$8,221,559	\$37,727	\$430,262	\$8,689,548	\$606,728	\$2,237,004	\$6,465	\$172,760	\$3,022,957	\$5,666,590
<i>15</i>	\$0	\$8,627,426	\$39,613	\$443,170	\$9,110,209	\$645,212	\$1,889,277	\$6,788	\$177,943	\$2,719,220	\$6,390,990
<i>16</i>	\$0	\$9,050,246	\$41,594	\$456,465	\$9,548,305	\$685,458	\$1,502,274	\$7,128	\$183,281	\$2,378,140	\$7,170,165
<i>17</i>	\$0	\$9,490,671	\$43,674	\$470,159	\$10,004,504	\$727,538	\$1,073,458	\$7,484	\$188,779	\$1,997,260	\$8,007,244
<i>18</i>	\$0	\$9,949,376	\$45,857	\$484,264	\$10,479,497	\$771,526	\$600,163	\$7,858	\$194,443	\$1,573,990	\$8,905,507
<i>19</i>	\$0	\$10,427,062	\$48,150	\$498,792	\$10,974,004	\$817,499	\$627,128	\$8,251	\$200,276	\$1,653,154	\$9,320,849
<i>20</i>	\$0	\$10,924,454	\$50,558	\$513,755	\$11,488,767	\$865,536	\$655,171	\$8,664	\$206,284	\$1,735,655	\$9,753,112

Table 28: Estimation of Interest Loss from Users

Year	Traffic Vol.	ETC (%)	ETC Transaction	Payment Split among ETC Users		ETC Transaction		Number of ETC Account ¹		Total Balance ²		Annual Interest*
				Cash & Checks	Credit Card	Cash & Checks	Credit Card	Cash & Checks	Credit Card	Cash & Checks	Credit Card	
<i>Base</i>	19,016,173	0%	0	64%	36%	0	0	0	0	\$0	\$0	\$0
<i>1</i>	19,565,364	6%	1,173,922	64%	36%	751,310	422,612	4,696	2,641	\$284,089	\$52,826	\$20,215
<i>2</i>	20,114,556	15%	3,017,183	64%	36%	1,930,997	1,086,186	12,069	6,789	\$730,158	\$135,773	\$51,956
<i>3</i>	20,663,747	20%	4,132,749	64%	36%	2,644,960	1,487,790	16,531	9,299	\$1,000,125	\$185,974	\$71,166
<i>4</i>	21,212,938	25%	5,303,235	64%	36%	3,394,070	1,909,164	21,213	11,932	\$1,283,383	\$238,646	\$91,322
<i>5</i>	21,762,130	30%	6,528,639	64%	36%	4,178,329	2,350,310	26,115	14,689	\$1,579,931	\$293,789	\$112,423
<i>6</i>	22,311,321	35%	7,808,962	64%	36%	4,997,736	2,811,226	31,236	17,570	\$1,889,769	\$351,403	\$134,470
<i>7</i>	22,860,512	40%	9,144,205	64%	36%	5,852,291	3,291,914	36,577	20,574	\$2,212,898	\$411,489	\$157,463
<i>8</i>	23,409,704	45%	10,534,367	64%	36%	6,741,995	3,792,372	42,137	23,702	\$2,549,317	\$474,047	\$181,402
<i>9</i>	23,958,895	50%	11,979,448	64%	36%	7,666,846	4,312,601	47,918	26,954	\$2,899,026	\$539,075	\$206,286
<i>10</i>	24,508,087	55%	13,479,448	64%	36%	8,626,846	4,852,601	53,918	30,329	\$3,262,026	\$606,575	\$232,116
<i>11</i>	25,057,278	60%	15,034,367	64%	36%	9,621,995	5,412,372	60,137	33,827	\$3,638,317	\$676,547	\$258,892
<i>12</i>	25,606,469	65%	16,644,205	64%	36%	10,652,291	5,991,914	66,577	37,449	\$4,027,898	\$748,989	\$286,613
<i>13</i>	26,155,661	70%	18,308,962	64%	36%	11,717,736	6,591,226	73,236	41,195	\$4,430,769	\$823,903	\$315,280
<i>14</i>	26,704,852	75%	20,028,639	64%	36%	12,818,329	7,210,310	80,115	45,064	\$4,846,931	\$901,289	\$344,893
<i>15</i>	27,254,043	80%	21,803,235	64%	36%	13,954,070	7,849,164	87,213	49,057	\$5,276,383	\$981,146	\$375,452
<i>16</i>	27,803,235	85%	23,632,749	64%	36%	15,124,960	8,507,790	94,531	53,174	\$5,719,125	\$1,063,474	\$406,956
<i>17</i>	28,352,426	90%	25,517,183	64%	36%	16,330,997	9,186,186	102,069	57,414	\$6,175,158	\$1,148,273	\$439,406
<i>18</i>	28,901,617	95%	27,456,536	64%	36%	17,572,183	9,884,353	109,826	61,777	\$6,644,482	\$1,235,544	\$472,802
<i>19</i>	29,450,809	95%	27,978,268	64%	36%	17,906,092	10,072,177	111,913	62,951	\$6,770,741	\$1,259,022	\$481,786
<i>20</i>	30,000,000	95%	28,500,000	64%	36%	18,240,000	10,260,000	114,000	64,125	\$6,897,000	\$1,282,500	\$490,770



Time Value

The main benefit gained by road users is time saving after introduction of ETC system. The amount of time is saved from acceleration, deceleration process and reduced delay (Idling event). On ETC lane, users don't need to slow down from the normal travel speed to pay the toll, waiting for transaction processing (Idling event), and accelerate back to normal travel speed.

Under the ETC scenario, there is still certain amount of manual users. Expected ETC split and annual traffic volume drew a nice picture about the number of manual users and ETC users per year. To estimate the time saved from deceleration and acceleration process, constant acceleration/deceleration rate is assumed. Moreover, the queue is assumed not to occupy any physical space. You can imagine that a driver starts to take brake at a certain distance prior to toll plaza and stops completely right in front of the toll plaza. If someone else is paying toll when he arrives, he waits on the top of the one who arrives earlier than him, like a pile of stack right behind the toll plaza. The average deceleration/acceleration distance is captured by the term “distance prior to (leading from) the toll plaza on “Travel at the Toll Plaza” on “*Input-Start*”.

Assume that prevailing travel speed on adjacent highway section is 55 mph as input, 0.2 miles of ramp prior to toll plaza as well as leading from toll plaza. Along with constant deceleration/acceleration rate, the amount of time saving from acceleration/deceleration process is shown on. Time saving from the idling event is the amount of reduced random and stop delay estimated on “Delay”. Here comes the question that how valuable this amount of time is?

In order to assign the time value to each unit of time, we have to know the mode split, average vehicle occupancy of each mode, and the unit time value of a user for each mode. From the chart “Travel at the Toll Plaza” on the sheet “*Input-Start*”, we obtain these entries. Next step is to calculate the weighted value of a unit time in year n: F_n .

$$F_n = (1 + I_f)^n \cdot \sum_k U_k \cdot VOC_k \cdot SU_k$$

where

- I_f : inflation rate
- U_k : value of time per hour for passenger using mode k in base year.
- VOC_k : average vehicle occupancy for mode k in base year.
- SU_k : mode split among ETC users for mode k in base year.

Then, the value of time saving is obtained by multiplying total amount of time saving by the weighted value, as shown on Table 30.



Time Value

The main benefit gained by road users is time saving after introduction of ETC system. The amount of time is saved from acceleration, deceleration process and reduced delay (Idling event). On ETC lane, users don't need to slow down from the normal travel speed to pay the toll, waiting for transaction processing (Idling event), and accelerate back to normal travel speed.

Under the ETC scenario, there is still certain amount of manual users. Expected ETC split and annual traffic volume drew a nice picture about the number of manual users and ETC users per year. To estimate the time saved from deceleration and acceleration process, constant acceleration/deceleration rate is assumed. Moreover, the queue is assumed not to occupy any physical space. You can imagine that a driver starts to take brake at a certain distance prior to toll plaza and stops completely right in front of the toll plaza. If someone else is paying toll when he arrives, he waits on the top of the one who arrives earlier than him, like a pile of stack right behind the toll plaza. The average deceleration/acceleration distance is captured by the term “distance prior to (leading from) the toll plaza on “Travel at the Toll Plaza” on “*Input-Start*”.

Assume that prevailing travel speed on adjacent highway section is 55 mph as input, 0.2 miles of ramp prior to toll plaza as well as leading from toll plaza. Along with constant deceleration/acceleration rate, the amount of time saving from acceleration/deceleration process is shown on. Time saving from the idling event is the amount of reduced random and stop delay estimated on “Delay”. Here comes the question that how valuable this amount of time is?

In order to assign the time value to each unit of time, we have to know the mode split, average vehicle occupancy of each mode, and the unit time value of a user for each mode. From the chart “Travel at the Toll Plaza” on the sheet “*Input-Start*”, we obtain these entries. Next step is to calculate the weighted value of a unit time in year n: F_n .

$$F_n = (1 + I_f)^n \cdot \sum_k U_k \cdot VOC_k \cdot SU_k$$

where

I_f : inflation rate

U_k : value of time per hour for passenger using mode k in base year.

VOC_k : average vehicle occupancy for mode k in base year.

SU_k : mode split among ETC users for mode k in base year.

Then, the value of time saving is obtained by multiplying total amount of time saving by the weighted value, as shown on Table 30.

Table 29: Estimation of the Amount of Time Saving as a Result of Introducing ETC System

Year	Traffic Vol.	ETC (%)	ETC Traffic	Idling Time (hrs/yr) ²			Time Difference due to Deceleration (hrs/yr) ³			Time Difference due to Acceleration (hrs/yr) ⁴				Total Time Saving** (hrs/yr)	
				Baseline	ETC Alternative	Time Saving	Avg. Speed (mph)	Baseline	ETC Alternative	Time Saving	Avg. Speed (mph)	Baseline	ETC Alternative		Time Saving
Base	19,016,173	0%	0												
1	19,565,364	5%	929,355	10,667	13,489	-2,822	28	142,294	138,914	3,379	28	142,294	138,914	3,379	3,937
2	20,114,556	10%	1,910,883	10,113	10,479	-366	28	146,288	139,339	6,949	28	146,288	139,339	6,949	13,531
3	20,663,747	14%	2,944,584	10,145	11,300	-1,155	28	150,282	139,574	10,708	28	150,282	139,574	10,708	20,260
4	21,212,938	19%	4,030,458	10,194	10,705	-511	28	154,276	139,620	14,656	28	154,276	139,620	14,656	28,802
5	21,762,130	24%	5,168,506	10,262	9,723	538	28	158,270	139,475	18,795	28	158,270	139,475	18,795	38,128
6	22,311,321	29%	6,358,727	10,345	9,565	780	28	162,264	139,142	23,123	28	162,264	139,142	23,123	47,025
7	22,860,512	33%	7,601,120	10,442	9,112	1,330	28	166,258	138,618	27,640	28	166,258	138,618	27,640	56,610
8	23,409,704	38%	8,895,687	10,552	8,231	2,321	28	170,252	137,904	32,348	28	170,252	137,904	32,348	67,017
9	23,958,895	43%	10,242,428	10,674	8,113	2,561	28	174,247	137,001	37,245	28	174,247	137,001	37,245	77,051
10	24,508,087	48%	11,641,341	10,805	7,035	3,770	28	178,241	135,908	42,332	28	178,241	135,908	42,332	88,434
11	25,057,278	52%	13,092,428	10,945	6,547	4,398	28	182,235	134,626	47,609	28	182,235	134,626	47,609	99,616
12	25,606,469	57%	14,595,687	11,093	5,699	5,394	28	186,229	133,154	53,075	28	186,229	133,154	53,075	111,545
13	26,155,661	62%	16,151,120	11,248	5,153	6,095	28	190,223	131,492	58,731	28	190,223	131,492	58,731	123,557
14	26,704,852	67%	17,758,727	11,409	4,282	7,127	28	194,217	129,640	64,577	28	194,217	129,640	64,577	136,281
15	27,254,043	71%	19,418,506	11,575	3,548	8,027	28	198,211	127,598	70,613	28	198,211	127,598	70,613	149,252
16	27,803,235	76%	21,130,458	11,746	2,819	8,927	28	202,205	125,367	76,838	28	202,205	125,367	76,838	162,603
17	28,352,426	81%	22,894,584	11,921	2,041	9,880	28	206,199	122,946	83,253	28	206,199	122,946	83,253	176,386
18	28,901,617	86%	24,710,883	12,100	1,198	10,903	28	210,194	120,336	89,858	28	210,194	120,336	89,858	190,618
19	29,450,809	90%	26,579,355	12,343	1,247	11,097	28	214,188	117,535	96,652	28	214,188	117,535	96,652	204,401
20	30,000,000	95%	28,500,000	12,587	1,297	11,289	28	218,182	114,545	103,636	28	218,182	114,545	103,636	218,562

Table 30: The Value of Time Saving

Year	Traffic Vol.	ETC (%)	Total Time Saving (hrs)				Factor of Time Value ¹	Total Value of Time Saving*
			From Idling	Deceleration	Acceleration	Total (hrs)		
<i>Base</i>	19,016,173	0%						
<i>1</i>	19,565,364	6%	-2,822	3,379	3,379	3,937	\$25	\$97,151
<i>2</i>	20,114,556	15%	-366	6,949	6,949	13,531	\$25	\$343,913
<i>3</i>	20,663,747	20%	-1,155	10,708	10,708	20,260	\$26	\$530,378
<i>4</i>	21,212,938	25%	-511	14,656	14,656	28,802	\$27	\$776,598
<i>5</i>	21,762,130	30%	538	18,795	18,795	38,128	\$28	\$1,058,901
<i>6</i>	22,311,321	35%	780	23,123	23,123	47,025	\$29	\$1,345,198
<i>7</i>	22,860,512	40%	1,330	27,640	27,640	56,610	\$29	\$1,667,969
<i>8</i>	23,409,704	45%	2,321	32,348	32,348	67,017	\$30	\$2,033,817
<i>9</i>	23,958,895	50%	2,561	37,245	37,245	77,051	\$31	\$2,408,483
<i>10</i>	24,508,087	55%	3,770	42,332	42,332	88,434	\$32	\$2,847,233
<i>11</i>	25,057,278	60%	4,398	47,609	47,609	99,616	\$33	\$3,303,461
<i>12</i>	25,606,469	65%	5,394	53,075	53,075	111,545	\$34	\$3,810,007
<i>13</i>	26,155,661	70%	6,095	58,731	58,731	123,557	\$35	\$4,346,931
<i>14</i>	26,704,852	75%	7,127	64,577	64,577	136,281	\$36	\$4,938,419
<i>15</i>	27,254,043	80%	8,027	70,613	70,613	149,252	\$37	\$5,570,695
<i>16</i>	27,803,235	85%	8,927	76,838	76,838	162,603	\$38	\$6,251,062
<i>17</i>	28,352,426	90%	9,880	83,253	83,253	176,386	\$40	\$6,984,377
<i>18</i>	28,901,617	95%	10,903	89,858	89,858	190,618	\$41	\$7,774,363
<i>19</i>	29,450,809	95%	11,097	96,652	96,652	204,401	\$42	\$8,586,595
<i>20</i>	30,000,000	95%	11,289	103,636	103,636	218,562	\$43	\$9,456,913

The other source of benefit to users is cost reduction in fuel purchasing as a result of time saving. The benefit to community is also of interest. The benefit to community comes from the reduction in emission of toxic gas, which harms health of human beings a lot. The last sheet in the file is devoted to these two subjects.

Fuel & Emission

Cost reduction for users from fuel purchasing is estimated by the total amount of time saving from acceleration process times the product of average fuel consumption per hour and average fuel cost per gallon. The majority of fuel consumption occurs during the acceleration process; therefore, the fuel consumption during the rest of time is ignorant for the convenience of calculation.

From environmental point of view, emission of NO_x, HC, and CO is something we don't like. The emission rate of these three pollutants during the deceleration process is light; thus, neglected. The emission rate of NO_x during the idling event is also light according to Li's report. From the input sheet, we got emission rates for these three pollutant during either acceleration or idling process. The amount of pollution reduced can be estimated as follows:

- ✱ Emission Reduction of Pollutant p from Idling event in year n (ER_{pn}), in kgs.

$$ER_{pn} = T_{idle,n} * E_{idle,p} * 60 / 1000$$

where

$T_{idle,n}$: time saved in the idling event in year n , in hrs.

It is equal to differences in delay under baseline and ETC scenario.

$E_{idle,p}$: emission rate for pollutant type p during the idling event, in grams per minute.

- ✱ Emission Reduction of Pollutant type p from Acceleration Process in year n ($ER_{acc,p,n}$), in kgs.

$$ER_{acc,p,n} = (T_{abn} - T_{acn}) \cdot FG_a \cdot E_{acc,p} / 1000$$

where

$E_{acc,p}$: emission rate for pollutant type p during the acceleration, in grams per gallon.

Then, we multiply the reduced emission of each pollutant by the cost of each pollutant to estimate the value of the reduction of fuel reduction.

The results are shown on Table 31 Table 32.

And then, we are done.

Table 31: Estimation of Vehicle Emission (kgs)

Year	Traffic Vol.	ETC (%)	Total Time Saving per Year (hrs)				Emission Reduction from Idling (kgs) ¹			Emission Reduction from Acceleration (kgs) ²			Total Emission Reduction (kgs)**		
			Idling	Deceleration	Acceleration	Total	Nox	HC	CO	Nox	HC	CO	Nox	HC	CO
<i>Base</i>	19,016,173	0%													
<i>1</i>	19,565,364	6%	-2,822	3,379	3,379	3,937	0	-25	-423	184	71	1,554	184	45	1,131
<i>2</i>	20,114,556	15%	-366	6,949	6,949	13,531	0	-3	-55	378	145	3,195	378	142	3,140
<i>3</i>	20,663,747	20%	-1,155	10,708	10,708	20,260	0	-10	-173	582	224	4,923	582	213	4,750
<i>4</i>	21,212,938	25%	-511	14,656	14,656	28,802	0	-5	-77	796	306	6,739	796	302	6,662
<i>5</i>	21,762,130	30%	538	18,795	18,795	38,128	0	5	81	1,021	393	8,642	1,021	398	8,723
<i>6</i>	22,311,321	35%	780	23,123	23,123	47,025	0	7	117	1,256	483	10,632	1,256	490	10,749
<i>7</i>	22,860,512	40%	1,330	27,640	27,640	56,610	0	12	199	1,502	578	12,709	1,502	590	12,909
<i>8</i>	23,409,704	45%	2,321	32,348	32,348	67,017	0	21	348	1,758	676	14,874	1,758	697	15,222
<i>9</i>	23,958,895	50%	2,561	37,245	37,245	77,051	0	23	384	2,024	778	17,125	2,024	801	17,509
<i>10</i>	24,508,087	55%	3,770	42,332	42,332	88,434	0	34	565	2,300	885	19,464	2,300	919	20,030
<i>11</i>	25,057,278	60%	4,398	47,609	47,609	99,616	0	40	660	2,587	995	21,891	2,587	1,035	22,550
<i>12</i>	25,606,469	65%	5,394	53,075	53,075	111,545	0	49	809	2,884	1,109	24,404	2,884	1,158	25,213
<i>13</i>	26,155,661	70%	6,095	58,731	58,731	123,557	0	55	914	3,191	1,227	27,005	3,191	1,282	27,919
<i>14</i>	26,704,852	75%	7,127	64,577	64,577	136,281	0	64	1,069	3,509	1,350	29,693	3,509	1,414	30,762
<i>15</i>	27,254,043	80%	8,027	70,613	70,613	149,252	0	72	1,204	3,837	1,476	32,468	3,837	1,548	33,672
<i>16</i>	27,803,235	85%	8,927	76,838	76,838	162,603	0	80	1,339	4,175	1,606	35,330	4,175	1,686	36,669
<i>17</i>	28,352,426	90%	9,880	83,253	83,253	176,386	0	89	1,482	4,524	1,740	38,280	4,524	1,829	39,762
<i>18</i>	28,901,617	95%	10,903	89,858	89,858	190,618	0	98	1,635	4,883	1,878	41,317	4,883	1,976	42,952
<i>19</i>	29,450,809	95%	11,097	96,652	96,652	204,401	0	100	1,664	5,252	2,020	44,441	5,252	2,120	46,105
<i>20</i>	30,000,000	95%	11,289	103,636	103,636	218,562	0	102	1,693	5,632	2,166	47,652	5,632	2,268	49,345

Table 32: Valuation of Emission Reduction and Fuel Consumption

Year	Total Emission Reduction (kgs)			Value of Emission Reduction				Value of Fuel Reduction
	Nox	HC	CO	Nox	HC	CO	Total	
<i>Base</i>								
<i>1</i>	184	45	1,131	\$234	\$58	\$7	\$299	\$8,178
<i>2</i>	378	142	3,140	\$481	\$181	\$20	\$682	\$16,816
<i>3</i>	582	213	4,750	\$742	\$272	\$30	\$1,044	\$25,912
<i>4</i>	796	302	6,662	\$1,015	\$385	\$42	\$1,442	\$35,468
<i>5</i>	1,021	398	8,723	\$1,302	\$507	\$55	\$1,864	\$45,483
<i>6</i>	1,256	490	10,749	\$1,602	\$625	\$68	\$2,295	\$55,957
<i>7</i>	1,502	590	12,909	\$1,915	\$752	\$81	\$2,748	\$66,890
<i>8</i>	1,758	697	15,222	\$2,241	\$889	\$96	\$3,226	\$78,282
<i>9</i>	2,024	801	17,509	\$2,580	\$1,022	\$110	\$3,713	\$90,133
<i>10</i>	2,300	919	20,030	\$2,933	\$1,171	\$126	\$4,230	\$102,444
<i>11</i>	2,587	1,035	22,550	\$3,299	\$1,319	\$142	\$4,760	\$115,213
<i>12</i>	2,884	1,158	25,213	\$3,677	\$1,476	\$159	\$5,312	\$128,442
<i>13</i>	3,191	1,282	27,919	\$4,069	\$1,635	\$176	\$5,880	\$142,130
<i>14</i>	3,509	1,414	30,762	\$4,474	\$1,803	\$194	\$6,471	\$156,277
<i>15</i>	3,837	1,548	33,672	\$4,892	\$1,974	\$212	\$7,078	\$170,883
<i>16</i>	4,175	1,686	36,669	\$5,324	\$2,150	\$231	\$7,705	\$185,948
<i>17</i>	4,524	1,829	39,762	\$5,768	\$2,332	\$250	\$8,350	\$201,472
<i>18</i>	4,883	1,976	42,952	\$6,226	\$2,520	\$271	\$9,016	\$217,456
<i>19</i>	5,252	2,120	46,105	\$6,696	\$2,703	\$290	\$9,690	\$233,898
<i>20</i>	5,632	2,268	49,345	\$7,180	\$2,891	\$311	\$10,382	\$250,800